Genome-scale context, stoichiometric refinement, and neuro-symbolic reasoning for proteome function annotation

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Abstract

Protein function prediction methods typically score sequences or structures one protein at a time, whereas long-read sequencing and modern genome assembly now routinely recover complete chromosomes that expose operon organization, macromolecular complex stoichiometry, metabolic network connectivity, and clade-specific anatomical constraints across the entire proteome. Here we present Genome-Scale Protein Annotation (GSPA), a four-stage framework that calibrates and pools complementary per-protein predictors (including our per-protein meta-integrator, GSPA-Seq), refines proteome-wide posteriors across co-directional operons, essential functions, pathways, obligate heteromeric complexes, and pre-propagation taxonomic and anatomical ontology axioms, and couples reaction-local genomic context with positive-cost parsimonious flux balance analysis, competitive gene-locus attribution, and selective ontology-guarded language-model reasoning. In a whole-genome Critical Assessment of Functional Annotation (CAFA) evaluation that withholds six complete reference genomes (20,697 coding sequences) across prospective time-based splits and sequence-similarity homology splits ($\leq$ 100%, $\leq$ 70%, $\leq$ 50%, and $\leq$ 30% identity) and compares against both official pre-trained and from-scratch retrained external predictors (DeepFRI, DeepGraphGO, TALE+, SPROF-GO, DeepGO-SE, DPFunc, 16-kernel DeepGOCNN/DeepGOPlus, and 3-billion-parameter ESM-2 3B architectures), multi-modal per-protein integration in GSPA-Seq outperforms every external baseline, and whole-genome inference in GSPA improves information-accretion-weighted f_w and $F_{\max}$ across all six splits ($f_w = 0.4996$, $F_{\max} = 0.5490$ on prospective targets with pre-t_0 chromosomal context, a +0.0541 f_w gain over completely withheld chromosomes at 0.4455; and $f_w = 0.5055$, $F_{\max} = 0.5272$ at $\leq$ 30% identity, where sequence alignment falls to $f_w = 0.1163$ and retrained external baselines reach 0.2702–0.3509). Operon context, pathway and complex coherence, flux-certified metabolic gap-filling, and pre-propagation ontology constraint enforcement each contribute positively to CAFA accuracy while eliminating 100% of taxonomic and anatomical contradictions. Across bacteria, archaea, viruses, and eukaryotes, semantic ancestor relaxation outperforms term deletion in multicellular animals and vascular plants, operon-coupled complex coherence raises recovery of held-out heteromeric complex subunits (Hit@1) from 56.0% to 93.7%, positive-cost stoichiometric refinement eliminates the 83.7% zero-flux reaction padding of negative-cost gap-filling and improves identification of fitness-validated bacterial enzymes and experimental BacDive phenotypes across unseen genera, and selective guarded reasoning on fewer than 2.5% of proteins raises top-ranked recovery of ambiguous missing pathway enzymes from 34.0% to 82.7%. Applied to 72,533 complete genomes from AllTheBacteria (292.1 million coding sequences across 11,632 species and 37 phyla), GSPA rescues 42.9% (51.4 million) of sequence dark-matter proteins ($<$ 30% identity, 76.4% lacking any detectable BLAST alignment) and 77.0% (60.7 million) of twilight-zone proteins while eliminating 2.44 billion cross-kingdom ontology violations, revealing a genome-size bifurcation between drift-driven core dark matter in streamlined bacteria and a 136.4-fold metabolic gap surge in large environmental genomes, separating 12.51 million obligate complex subunits from 1.43 million domain-reuse singletons, and mapping non-orthologous gene displacement across bacterial pathways and complexes.

1 Introduction

Automated protein function annotation assigns biological roles to the protein-coding genes of an organism and forms the basis of comparative genomics, metabolic modeling, and systems biology [1, 2]. We consider the genome-scale function inference problem in which the input consists of the complete set of protein-coding genes $\mathcal{P} = \{p_1, \ldots, p_N\}$ of an organism, their chromosomal coordinates and strand organization $\mathcal{K}$, the organism's taxonomic lineage τ_{tax}, and reference biochemical and ontological knowledge bases, and the output is a joint assignment of Gene Ontology (GO) terms [3] and Enzyme Commission (EC) numbers across $\mathcal{P}$. Because the proteins encoded by a genome evolve and

act together within a single cell or organism, a valid joint assignment satisfies structural invariants that span many proteins simultaneously: core cellular functions required in the organism's domain of life are represented, metabolic pathways form stoichiometrically connected routes that support biomass synthesis [4, 5], obligate heteromeric complexes are annotated with their complete partner subunit sets [6, 7], and assigned terms respect both the organism's taxonomic lineage and the anatomical body plan of its clade [8, 9]. These organism-level invariants link the function of each protein to the architecture of the complete genome.

Formulating function prediction over complete genomes is timely because chromosome-scale assemblies are now broadly accessible. Historically, short-read sequencing of microbial isolates and metagenomes yielded fragmented contigs from which open reading frames were predicted and annotated as isolated sequences, severing operons, biosynthetic gene clusters, and chromosome-wide reaction inventories. Advances in long-read sequencing [10, 11] and repeat-graph genome and metagenome assemblers [12] routinely recover complete or near-complete chromosomes for both cultivated isolates and metagenome-assembled genomes (MAGs) of uncultured lineages. Chromosomal gene order, co-directional operons, and complete proteome complements are therefore standard inputs that function prediction methods can exploit directly.

Standard function predictors derive representations of an individual protein sequence or structure and transfer annotations from reference databases. Sequence alignment transfers annotations from homologous sequences [13–15]; profile hidden Markov models and domain signatures identify conserved functional motifs [16, 17]; deep convolutional, transformer, and graph neural networks capture sequence motifs, domain graphs, and GO label hierarchies [18–21]; protein language models and semantic entailment embeddings project sequences into representation spaces where functionally related proteins cluster [22–26]; and structural graph networks transfer functions across remote fold similarities [27–29]. Because these modalities capture complementary biochemical properties, multi-source ensembles such as GOLabeler [30] and NetGO [31, 32], as well as our per-protein multi-modal stacker GSPA-Seq, achieve strong protein-centric accuracy in Critical Assessment of Functional Annotation (CAFA) benchmarks [1, 2, 33]. However, scoring each protein p_i in isolation leaves four sources of genome-scale information unused: co-transcribed operon neighbors that share macromolecular complexes or metabolic pathways, partially annotated chromosomes where previously characterized genes can anchor newly studied neighbors, proteome-wide stoichiometric and complex completeness requirements, and clade-specific taxonomic and anatomical constraints that can prevent false-positive leaf predictions from propagating upward through the GO hierarchy.

Comparative genomics, constraint-based metabolic modeling, and formal bio-ontologies each address part of this organism-level context. On the chromosome, conserved gene order, co-directional operons, and genomic language models [34–41] capture local synteny within contiguous gene windows, while domain-essential profiles and dispersed complex stoichiometry act across the full proteome. In systems biology, metabolic reconstruction pipelines including Pathway Tools [42], ModelSEED [43], CarveMe [44], gapseq [45], and GapMind [46] identify reactions required for steady-state flux, and our earlier method METEOR [47] coupled metabolic gap-filling with neural enzyme prediction via reaction-indicator mixed-integer linear programming (MILP). Inspection of reaction-indicator MILP reveals two mathematical properties that limit gene-level precision: negative objective costs on candidate reactions above an addition threshold reward the solver for adding reactions with zero steady-state flux, and uniform reaction-to-gene boosting elevates paralogs equally without modeling operon neighborhoods, subunit roles, or competition among candidate loci. In bio-ontologies, GO taxon rules [8] and cross-ontology axioms linking GO processes to the Uberon anatomy ontology [9, 48], the Plant Ontology (PO) [49], and the Cell Ontology (CL) [50] define which structures and processes can exist in a clade; enforcing these rules before upward GO propagation and relaxing forbidden terms to their nearest valid ancestors preserves conserved ancestral functions while preventing cross-clade errors. Finally, for the small fraction of loci that remain ambiguous after deterministic probabilistic and stoichiometric refinement, tool-using language-model agents [51, 52] can synthesize heterogeneous domain, operon, and pathway evidence when verified by a symbolic ontology guardrail.

Here we present Genome-Scale Protein Annotation (GSPA), a four-stage framework that unifies per-protein representation learning with whole-genome probabilistic, stoichiometric, and ontological inference (Figure 1). First, Stage 0 (together with our per-protein multi-modal integrator GSPA-Seq) calibrates and pools sequence homology, domain signatures, 16-kernel 1D convolutional networks, and 3-billion-parameter protein language models (`ESM-2 3B`) to establish a strong per-protein prior. Second, Stage 1 and Stage 2A exploit whole-genome organization: Stage 1 iteratively updates proteome-wide log-odds across essential functions (B_1), pathways and obligate heteromeric complexes (B_2), taxonomic consistency (B_3), cross-ontology anatomy (B_4), metabolic gaps (B_5), and co-directional operons (B_6), and Stage 2A enforces taxonomic and anatomical constraints prior to upward GO propagation while relaxing clade-incompatible terms to their nearest valid ancestors. Third, Stage 1, Stage 2B, and Stage 3 extend inference beyond sequence homology: Stage 1 localizes missing pathway enzymes via reaction-local genomic context (RLGC), Stage 2B couples role-aware disjunctive normal form (DNF) aggregation of isozymes and subunits with positive-cost lexicographic parsimonious flux balance analysis (pFBA) [53] and competitive locus attribution against an unassigned-

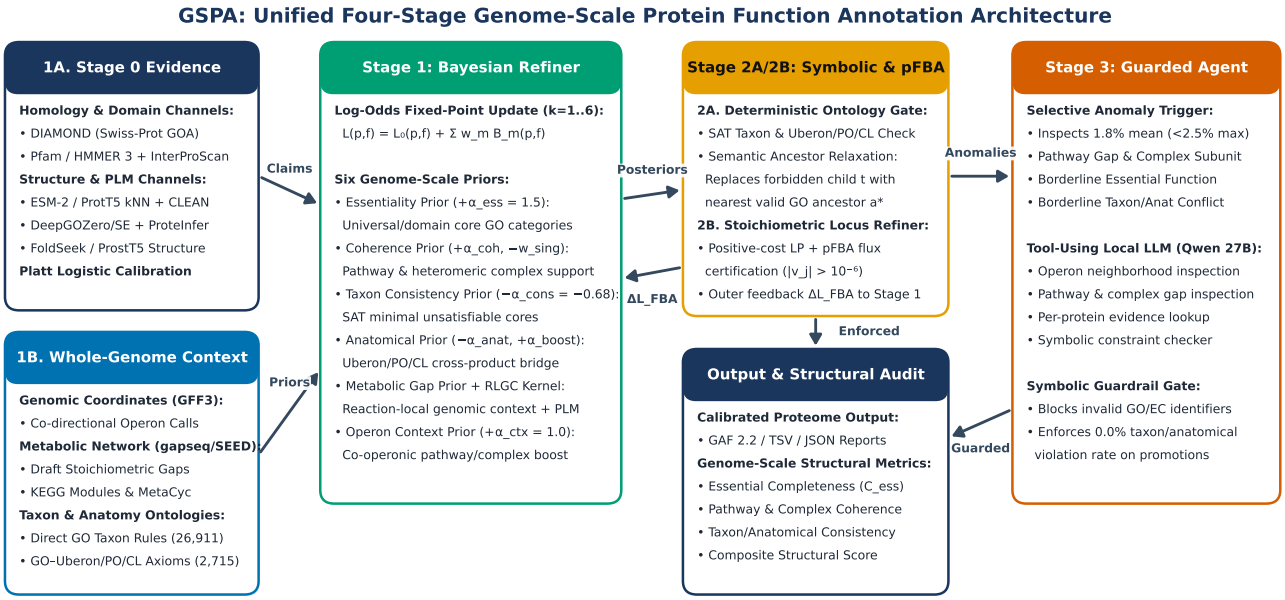


Figure 1. Four-stage architecture of GSPA. Stage 0 calibrates per-channel confidence scores from sequence homology, domain profiles, protein language models, and structural similarity (or our per-protein multi-modal stacker GSPA-Seq) and pools them via correlation-grouped Noisy-OR into an initial proteome-wide probability matrix. Stage 1 refines these probabilities in log-odds space using six whole-proteome priors covering core essentiality (B_1), pathway and heteromeric complex coherence (B_2), taxonomic consistency (B_3), cross-ontology anatomical constraints (B_4, 26,911 direct GO taxon rules and 2,715 GO–Uberon/PO/CL axioms), metabolic gaps coupled to reaction-local genomic context (B_5, RLGC), and co-directional operons (B_6). Stage 2A enforces symbolic ontology constraints prior to final ancestor propagation with semantic ancestor relaxation and prunes isolated obligate complex singletons; Stage 2B aggregates isozyme and subunit roles in disjunctive normal form (DNF), repairs the metabolic network via positive-cost lexicographic parsimonious flux balance analysis (pFBA), and attributes flux-certified reactions competitively to candidate gene loci against an unassigned-gene hypothesis $h_\varnothing$. Stage 3 applies selective tool-mediated language-model reasoning to fewer than 2.5% of proteins under a deterministic symbolic ontology guardrail.

gene hypothesis, and Stage 3 resolves residual ambiguities on fewer than 2.5% of proteins using a tool-calling language model verified by a symbolic ontology guardrail. Across eight benchmark suites spanning bacteria, archaea, viruses, and eukaryotes (Supplementary Table S1) and a pan-phylum application to 72,533 complete genomes from AllTheBacteria [54], we test six hypotheses: (i) whether withholding complete genomes and conditioning on chromosomal operons, pathway and complex coherence, metabolic gap-filling and FBA, and pre-propagation ontology constraints improves protein-centric CAFA accuracy over pre-trained and retrained external baselines across both time-based and homology-based splits; (ii) whether whole-genome priors improve organism-level structural coherence while preserving protein-level accuracy; (iii) whether operon architecture and cross-ontology axioms complete heteromeric complexes and repair eukaryotic anatomical errors; (iv) whether positive-cost flux modeling, competitive locus attribution, and pathway coupling eliminate zero-flux reaction padding and improve fitness-validated enzyme and experimental phenotype prediction; (v) when selective guarded language-model reasoning complements deterministic and supervised inference; and (vi) what macro-evolutionary principles govern sequence dark matter, genome-size scaling, heteromeric complex stoichiometry, and non-orthologous gene displacement across bacterial phyla. GSPA v2.1.0, precomputed ontology bridges, and benchmark workflows are available at https://github.com/bio-ontology-research-group/gspa.

2 Results

2.1 Whole-genome inference improves CAFA function prediction across time-based and homology-based splits

We first asked whether calibrating and pooling complementary per-protein predictors yields a stronger baseline annotation than individual predictors and production genome annotation pipelines when evaluated under an identical ancestor-propagated protocol. We compared GSPA against its individual sequence alignment, domain profile, protein language model, and structural similarity channels, against NCBI PGAP [14], InterProScan [17], and metagenomic-deepFRI [55], and against the genomic language models gLM and gLM2 [38, 39] across 13-genome and 21-genome reference panels (Figure 2d, Table 2, and Supplementary Table S2), with all pipeline hyperparameters calibrated on *Escherichia coli* K-12 and held frozen across all out-of-sample species (Supplementary Table S3). Across the 11 prokaryotic proteomes of the 13-genome panel, GSPA with three sequence and domain channels reaches a micro $F_{\max}$ of 0.8421, exceeding NCBI PGAP (0.7820). Extending Stage 0 to all seven neural, language-model, and struc-

Table 1. Whole-genome withheld CAFA evaluation across prospective time-based splits (`WG_Time_Rollover`, `WG_Time_Withheld`) and sequence-similarity homology splits (`WG_Homology_le100`, `WG_Homology_le70`, `WG_Homology_le50`, `WG_Homology_le30`) across six complete reference proteomes (20,697 **ORFs,** 3,809 **operons,** 1,236 **metabolic gaps), comparing against official pre-trained and from-scratch retrained external predictors.** Six complete prokaryotic genomes (*B. subtilis* 168, *P. aeruginosa* PAO1, *M. tuberculosis* H37Rv, *H. pylori* 26695, *Synechocystis* sp. PCC 6803, and *M. jannaschii* DSM 2661) are withheld as complete genome inputs (20,697 coding sequences, 3,315 experimentally annotated target proteins, and 1,060 strictly post-t_0 target proteins). Official pre-trained weights of DeepFRI and DeepGO-SE are evaluated on the time-based splits, and all trainable baselines (DeepFRI GCN + LM, DeepGraphGO, TALE+, SPROF-GO, DeepGO-SE with $K = 5$ ELEmbeddings world models, DPFunc, 16-kernel DeepGOCNN and DeepGOPlus, ESM-2 3B k-NN/MLP/C-HMCNN, InterPro-LR, and out-of-fold GSPA-Seq meta-stacker) are retrained from scratch on each split's filtered training corpus (complete 130-row evaluation across all 22 methods in Supplementary Table S4). Columns report official CAFA information-accretion-weighted f_w across Molecular Function (MF), Biological Process (BP), Cellular Component (CC), and aspect mean, unweighted protein-centric $F_{\max}$ aspect mean, and total surviving formal taxonomic and anatomical constraint violations (V) at posterior ≥ 0.50.

Whole-Genome Split ($N_{\text{train}} \rightarrow N_{\text{eval}}$; 20,697 ORFs)	Pre-trained or Retrained Baseline / Whole-Genome Ablation	f_w (MF)	f_w (BP)	f_w (CC)	Mean f_w	$F_{\max}$ (MF / BP / CC)	Mean $F_{\max}$	Violations (V)
WG_Time_Rollover Prospective $t_0 \rightarrow t_1$ w/ Pre-t_0 Chromosomal Context (82,404 → 1,060 Targets)	DeepFRI (Pretrained Official)	0.2655	0.1705	0.2886	0.2415	0.2779/0.2217/0.3642	0.2879	1,300
	DIAMOND BLAST-kNN (CAFA Baseline 2)	0.4829	0.3361	0.3070	0.3753	0.4929/0.3777/0.3487	0.4064	988
	DeepFRI (GCN + LM, Retrained)	0.4408	0.2752	0.4366	0.3842	0.4648/0.3535/0.5678	0.4620	61
	DeepGOPlus (16-Kernel CNN + DIAMOND-kNN, Retrained)	0.5435	0.3341	0.3040	0.3939	0.5615/0.3896/0.4258	0.4590	489
	DeepGO-SE (ESM-2 3B + ELEmbeddings, Retrained)	0.5068	0.2799	0.4944	0.4270	0.5255/0.3588/0.6108	0.4984	83
	DeepGO-SE (Pretrained Official)	0.5077	0.3014	0.5491	0.4527	0.5264/0.3102/**0.6200**	0.4855	113
	ESM-2 3B C-HMCNN (Hierarchical MCLoss, Retrained)	0.5095	0.3362	0.5141	0.4533	0.5398/0.3975/0.6186	0.5186	237
	TALE+ (Transformer + GO Label Attn + DIAMOND, Retrained)	0.5435	0.3584	0.4624	0.4548	0.5561/0.4236/0.5807	0.5202	216
	DeepGraphGO (InterPro + GraphGCN, Retrained)	0.5051	0.3353	0.5249	0.4551	0.5344/0.3964/0.6180	0.5162	79
	DPFunc (Domain Cross-Attn + GCN, Retrained)	0.5143	0.3476	0.5279	0.4633	0.5364/0.4148/0.6172	0.5228	318
	SPROF-GO (PLM + Hierarchy + Label Diffusion, Retrained)	0.5771	0.3691	0.5122	0.4861	0.5928/0.4309/0.6163	0.5467	99
	GSPA-Seq (Ours, Per-Protein Integrator)	0.5737	0.3940	0.5047	0.4908	0.5891/**0.4479**/0.5985	0.5452	670
	GSPA + Operon Genomic Context Only (B_6)	0.5800	0.3948	0.5082	0.4943	0.5936/0.4475/0.5994	0.5469	670
	GSPA + Pathway & Complex Coherence Only (B_1, B_2)	0.5785	0.3940	0.5055	0.4927	0.5917/0.4473/0.5989	0.5460	670
	GSPA + Taxon & Anatomical Constraints Only ($B_3, B_4 + 2A$)	0.5737	0.3940	0.5162	0.4946	0.5891/**0.4479**/0.6030	0.5467	0
	GSPA Full Whole-Genome (Stages 0 + 1 + 2A + 2B)	0.5826	0.3947	0.5214	0.4996	0.5956 /0.4473/0.6042	0.5490	0
WG_Time_Withheld Prospective $t_0 \rightarrow t_1$ w/ All 6 Genomes Purged from Train (79,880 → 1,060 Targets)	DeepFRI (GCN + LM, Retrained)	0.4257	0.2592	0.2974	0.3274	0.4618/0.3463/0.4568	0.4216	308
	DIAMOND BLAST-kNN (CAFA Baseline 2)	0.4581	0.3022	0.2345	0.3316	0.4699/0.3465/0.2907	0.3690	1,195
	DeepGO-SE (ESM-2 3B + ELEmbeddings, Retrained)	0.4789	0.2588	0.3560	0.3646	0.5002/0.3392/0.4846	0.4413	257
	DPFunc (Domain Cross-Attn + GCN, Retrained)	0.5034	0.3117	0.3599	0.3917	0.5282/0.3801/0.4917	0.4666	631
	TALE+ (Transformer + GO Label Attn + DIAMOND, Retrained)	0.5411	0.3380	0.3085	0.3959	0.5450/0.4034/0.4748	0.4744	371
	SPROF-GO (PLM + Hierarchy + Label Diffusion, Retrained)	0.5272	0.3378	0.3277	0.3976	0.5442/0.4059/0.4736	0.4746	655
	DeepGraphGO (InterPro + GraphGCN, Retrained)	0.5080	0.3145	0.3716	0.3980	0.5272/0.3838/**0.5149**	0.4753	200
	GSPA-Seq (Ours, Per-Protein Integrator)	0.5609	0.3680	0.3766	0.4352	0.5679/**0.4232**/0.4653	0.4854	1,219
	GSPA + Pathway & Complex Coherence Only (B_1, B_2)	0.5646	0.3688	0.3748	0.4361	0.5736/0.4231/0.4653	0.4873	1,219
	GSPA + Taxon & Anatomical Constraints Only ($B_3, B_4 + 2A$)	0.5609	0.3680	0.4021	0.4437	0.5679/**0.4232**/0.4675	0.4862	0
	GSPA Full Whole-Genome (Stages 0 + 1 + 2A + 2B)	0.5684	0.3653	0.4030	0.4455	0.5776 /0.4198/0.4675	0.4883	0
WG_Homology_le100 Leave-Whole-Genome-Out: All 6 Genomes Withheld (79,880 → 3,315 Targets)	DeepFRI (GCN + LM, Retrained)	0.4214	0.2951	0.2951	0.3372	0.4690/0.3700/0.4448	0.4279	843
	DIAMOND BLAST-kNN (CAFA Baseline 2)	0.4922	0.3505	0.2139	0.3522	0.5093/0.3884/0.2833	0.3937	4,340
	DeepGO-SE (ESM-2 3B + ELEmbeddings, Retrained)	0.4926	0.3020	0.3128	0.3691	0.5289/0.3692/0.4595	0.4525	730
	DPFunc (Domain Cross-Attn + GCN, Retrained)	0.5264	0.3460	0.3105	0.3943	0.5622/0.4123/0.4516	0.4754	1,982
	DeepGraphGO (InterPro + GraphGCN, Retrained)	0.5270	0.3508	0.3321	0.4033	0.5620/0.4153/**0.4769**	0.4847	373
	SPROF-GO (PLM + Hierarchy + Label Diffusion, Retrained)	0.5501	0.3878	0.3009	0.4129	0.5806/0.4482/0.4498	0.4929	823
	TALE+ (Transformer + GO Label Attn + DIAMOND, Retrained)	0.5684	0.3865	0.3156	0.4235	0.5849/0.4375/0.4562	0.4929	1,100
	GSPA-Seq (Ours, Per-Protein Integrator)	0.5920	0.4336	0.3381	0.4546	0.6083/0.4777/0.4503	0.5121	3,897
	GSPA + Pathway & Complex Coherence Only (B_1, B_2)	0.6075	0.4315	0.3365	0.4585	0.6230/0.4772/0.4503	0.5169	3,897
	GSPA + Taxon & Anatomical Constraints Only ($B_3, B_4 + 2A$)	0.5920	0.4341	0.3672	0.4645	0.6083/**0.4779**/0.4517	0.5126	0
	GSPA Full Whole-Genome (Stages 0 + 1 + 2A + 2B)	0.6076	0.4226	0.3754	0.4685	0.6232 /0.4731/0.4517	0.5160	0
WG_Homology_le70 All 6 Genomes + Homologs > 70% Identity Purged (79,457 → 3,315 Targets)	DeepFRI (GCN + LM, Retrained)	0.4098	0.2830	0.2686	0.3205	0.4595/0.3567/0.4092	0.4085	2,074
	DIAMOND BLAST-kNN (Filtered ≤ 70%)	0.4842	0.3473	0.2046	0.3454	0.5011/0.3853/0.2771	0.3878	4,355
	DeepGO-SE (ESM-2 3B + ELEmbeddings, Retrained)	0.4931	0.2998	0.3092	0.3674	0.5298/0.3697/0.4561	0.4519	558
	DPFunc (Domain Cross-Attn + GCN, Retrained)	0.5200	0.3443	0.3086	0.3910	0.5525/0.4139/0.4444	0.4703	1,208
	DeepGraphGO (InterPro + GraphGCN, Retrained)	0.5189	0.3498	0.3381	0.4023	0.5524/0.4114/**0.4843**	0.4827	514
	SPROF-GO (PLM + Hierarchy + Label Diffusion, Retrained)	0.5557	0.3869	0.3294	0.4240	0.5853/0.4465/0.4627	0.4982	1,858
	TALE+ (Transformer + GO Label Attn + DIAMOND, Retrained)	0.5662	0.3867	0.3223	0.4251	0.5857/0.4386/0.4583	0.4942	1,875
	GSPA-Seq (Ours, Per-Protein Integrator)	0.6143	0.4674	0.3480	0.4766	0.6242/0.5078/0.4631	0.5317	3,283
	GSPA + Pathway & Complex Coherence Only (B_1, B_2)	0.6198	0.4657	0.3475	0.4776	0.6311 /0.5068/0.4631	0.5337	3,283
	GSPA Full Whole-Genome (Stages 0 + 1 + 2A + 2B)	0.6179	0.4585	0.3895	0.4886	0.6300/**0.5025**/0.4645	0.5323	0
WG_Homology_le50 All 6 Genomes + Homologs > 50% Identity Purged (76,405 → 3,315 Targets)	DIAMOND BLAST-kNN (Filtered ≤ 50%)	0.4327	0.2930	0.1747	0.3001	0.4534/0.3331/0.2500	0.3455	6,231
	DeepFRI (GCN + LM, Retrained)	0.4208	0.2792	0.2976	0.3325	0.4701/0.3547/0.4126	0.4125	2,475
	DeepGO-SE (ESM-2 3B + ELEmbeddings, Retrained)	0.4843	0.2882	0.3149	0.3625	0.5254/0.3567/0.4615	0.4479	922
	DeepGraphGO (InterPro + GraphGCN, Retrained)	0.5012	0.3290	0.3169	0.3824	0.5372/0.3935/0.4593	0.4633	446
	DPFunc (Domain Cross-Attn + GCN, Retrained)	0.5041	0.3187	0.3431	0.3886	0.5406/0.3864/**0.4776**	0.4682	2,784
	TALE+ (Transformer + GO Label Attn + DIAMOND, Retrained)	0.5513	0.3523	0.3069	0.4035	0.5744/0.4093/0.4539	0.4792	1,761
	SPROF-GO (PLM + Hierarchy + Label Diffusion, Retrained)	0.5419	0.3567	0.3144	0.4043	0.5746/0.4193/0.4524	0.4821	1,458
	GSPA-Seq (Ours, Per-Protein Integrator)	0.6336	0.4678	0.3488	0.4834	0.6392/0.5086/0.4515	0.5331	3,443
	GSPA + Taxon & Anatomical Constraints Only ($B_3, B_4 + 2A$)	0.6338	0.4681	0.3857	0.4959	0.6393 /**0.5089**/0.4532	0.5338	0
	GSPA Full Whole-Genome (Stages 0 + 1 + 2A + 2B)	0.6283	0.4636	0.3985	0.4968	0.6350/0.5061/0.4532	0.5314	0
WG_Homology_le30 Twilight Zone: All 6 Genomes + Homologs > 30% Purged (61,323 → 3,315 Targets)	DIAMOND BLAST-kNN (Filtered ≤ 30%)	0.1756	0.1026	0.0707	0.1163	0.1872/0.1165/0.1065	0.1367	5,519
	DeepGOPlus (16-Kernel CNN + DIAMOND-kNN, Retrained)	0.3622	0.1892	0.2379	0.2631	0.4160/0.2430/0.3743	0.3444	4,433
	DeepFRI (GCN + LM, Retrained)	0.3660	0.2348	0.2099	0.2702	0.4280/0.2937/0.3827	0.3681	1,949
	DeepGO-SE (ESM-2 3B + ELEmbeddings, Retrained)	0.4124	0.2475	0.3246	0.3282	0.4652/0.3103/0.4345	0.4033	2,139
	DPFunc (Domain Cross-Attn + GCN, Retrained)	0.4235	0.2665	0.3031	0.3311	0.4729/0.3321/0.4434	0.4162	3,413
	DeepGraphGO (InterPro + GraphGCN, Retrained)	0.4201	0.2771	0.3162	0.3378	0.4690/0.3425/**0.4706**	0.4274	376
	SPROF-GO (PLM + Hierarchy + Label Diffusion, Retrained)	0.4389	0.2625	0.3291	0.3435	0.4868/0.3278/0.4499	0.4215	1,063
	TALE+ (Transformer + GO Label Attn + DIAMOND, Retrained)	0.4549	0.2466	0.3309	0.3442	0.4979/0.3088/0.4527	0.4198	2,787
	ESM-2 3B C-HMCNN (Hierarchical MCLoss, Retrained)	0.4360	0.2534	0.3633	0.3509	0.4822/0.3136/0.4615	0.4191	1,245
	GSPA-Seq (Ours, Per-Protein Integrator)	0.6399	0.4656	0.3630	0.4895	0.6447 /0.5031/0.4270	0.5249	2,756
	GSPA Full Whole-Genome (Stages 0 + 1 + 2A + 2B)	0.6391	0.4672	0.4100	0.5055	0.6446/**0.5039**/0.4330	0.5272	0

tural channels on the 21-genome panel raises micro $F_{\max}$ to 0.8705, outperforming the strongest single channel, ESM-2 nearest-neighbor transfer (0.8580). Combining complementary sequence and structural representations establishes a higher-accuracy starting distribution than any individual predictor, and we next tested whether withholding complete genomes and conditioning on whole-genome architecture improves CAFA prediction accuracy over pre-trained and retrained external baselines across both prospective time-based splits and sequence-similarity homology splits.

To test the central hypothesis that whole-genome inputs improve protein-centric CAFA function prediction, we designed a whole-genome withheld CAFA evaluation across six complete prokaryotic reference proteomes (*B. subtilis* 168, *P. aeruginosa* PAO1, *M. tuberculosis* H37Rv, *H. pylori* 26695, *Synechocystis* sp. PCC 6803, and *M. jannaschii* DSM 2661; 20,697 coding sequences, 3,315 experimentally annotated target proteins, 1,060 strictly post-t_0 newly annotated targets, 3,809 co-directional operons, and 1,236 metabolic gaps) spanning two prospective time-

based splits and four homology-based sequence-identity splits, comparing against official pre-trained weights of DeepFRI [27] and DeepGO-SE [25] on the temporal splits and retraining all 14 external and internal baseline models from scratch on each split's filtered training set (Table 1 and Supplementary Table S4). On the prospective time-based split with pre-t_0 chromosomal rollover context (`WG_Time_Rollover`, 82,404 training proteins evaluated on 1,060 post-t_0 targets across the 20,697 whole-genome ORFs), official pre-trained DeepFRI and DeepGO-SE reach mean information-accretion-weighted f_w values of 0.2415 ($F_{\max} = 0.2879$) and 0.4527 ($F_{\max} = 0.4855$), while baselines retrained from scratch on the t_0 corpus reach 0.3753 for DIAMOND BLAST-kNN, 0.3842 for DeepFRI ($F_{\max} = 0.4620$), 0.3939 for 16-kernel DeepGOPlus, 0.4270 for DeepGO-SE ($F_{\max} = 0.4984$), 0.4533 for ESM-2 3B C-HMCNN, 0.4548 for TALE+ [21] ($F_{\max} = 0.5202$), 0.4551 for DeepGraphGO [20] ($F_{\max} = 0.5162$), 0.4633 for DPFunc [29] ($F_{\max} = 0.5228$), and 0.4861 for SPROF-GO [24] ($F_{\max} = 0.5467$). Multi-channel per-protein integration reaches $f_w = 0.4882$ ($F_{\max} = 0.5389$) in GSPA Stage 0 and $f_w = 0.4908$ ($F_{\max} = 0.5452$, 670 ontology violations) in our per-protein integrator GSPA-Seq, while full whole-genome GSPA (Stages 0, 1, 2A, and 2B) reaches $f_w = 0.4996$ ($F_{\max} = 0.5490$, $S_{\min} = 9.63$) with zero ontology violations. When all 2,524 experimentally annotated proteins from the six species are purged from the training corpus (`WG_Time_Withheld`, 79,880 training proteins evaluated on the identical 1,060 post-t_0 targets), full GSPA again leads all retrained methods at $f_w = 0.4455$ ($F_{\max} = 0.4883$, 0 violations) compared with 0.3274 for DeepFRI, 0.3646 for DeepGO-SE, 0.3917 for DPFunc, 0.3959 for TALE+, 0.3976 for SPROF-GO, 0.3980 for DeepGraphGO, and 0.4352 for GSPA-Seq (1,219 violations). Comparing `WG_Time_Rollover` (0.4996) with `WG_Time_Withheld` (0.4455) on the same 1,060 post-t_0 proteins shows that pre-t_0 annotations on the surrounding chromosome provide a +0.0541 (+12.1%) gain in f_w. We next examined whether whole-genome inference preserves this advantage when cross-species sequence homologs are progressively removed and dissected the contribution of each genome-scale component.

Under the four whole-genome homology splits, where all six genomes plus all training proteins sharing $> 100\%$, $> 70\%$, $> 50\%$, or $> 30\%$ pairwise sequence identity with any protein in the six withheld genomes are purged prior to retraining (3,315 target proteins across 20,697 ORFs), DIAMOND BLAST-kNN degrades from $f_w = 0.3522$ at $\leq 100\%$ to 0.3454 at $\leq 70\%$, 0.3001 at $\leq 50\%$, and 0.1163 ($F_{\max} = 0.1367$) in the $\leq 30\%$ twilight zone, while retrained external baselines at $\leq 30\%$ identity drop to 0.2702 for DeepFRI, 0.3282 for DeepGO-SE, 0.3311 for DPFunc, 0.3378 for DeepGraphGO, 0.3435 for SPROF-GO, and 0.3442 for TALE+ (Table 1). Full whole-genome GSPA achieves the highest mean f_w across every homology regime: 0.4685 ($F_{\max} = 0.5160$) on leave-whole-genome-out `WG_Homology_le100`, 0.4886 ($F_{\max} = 0.5323$) at $\leq 70\%$, 0.4968 ($F_{\max} = 0.5314$) at $\leq 50\%$, and 0.5055 ($F_{\max} = 0.5272$) at $\leq 30\%$ identity, outperforming the strongest external baseline at $\leq 30\%$ identity (TALE+, $f_w = 0.3442$) by +0.1613 f_w and the per-protein GSPA-Seq integrator ($f_w = 0.4895$, 2,756 violations) by +0.0160 f_w while eliminating all 2,488–7,719 ontology violations (Table 1; multi-clade isolated-protein evaluation in Supplementary Table S15). Ablating each whole-genome module individually on top of GSPA-Seq confirms that all four genome-scale hypotheses contribute to CAFA accuracy through distinct GO aspects: (i) co-directional operon context (B_6) transfers heteromeric complex localization, high-information biological processes, and pathway steps across neighboring genes, raising mean f_w by +0.0035 on `WG_Time_Rollover` (0.4908 $\rightarrow$ 0.4943) and +0.0068 at $\leq 30\%$ identity (0.4895 $\rightarrow$ 0.4963); (ii) pathway, complex, and essentiality coherence (B_1, B_2) and metabolic gap-filling with FBA locus attribution (B_5 + Stage 2B) rescue missing catalytic steps and subunits, raising Molecular Function f_w by +0.0048 on `WG_Time_Rollover` (0.5737 $\rightarrow$ 0.5785) and +0.0155 on `WG_Homology_le100` (0.5920 $\rightarrow$ 0.6075); and (iii) enforcing taxonomic and anatomical constraints (B_3, B_4 +Stage 2A) prior to upward GO propagation suppresses forbidden eukaryotic organelle and developmental leaf terms before they can inflate shared intermediate ancestors, raising Cellular Component f_w by +0.0115 to +0.0372 across splits while reducing formal violations to zero.

We also evaluated enzyme EC prediction on a post-cutoff Swiss-Prot holdout of 125 enzymes (Supplementary Table S5) and across 69,666 proteins from 95 bacterial genomes under four donor-homology masking thresholds (Supplementary Table S6). On the post-cutoff holdout, pooling CLEAN [26], EnzBERT [56], and DeepProZyme [57] in Stage 0 reaches micro $F_{\max} = 0.4381$, and genome-scale refinement improves top-3 and top-5 enzyme recall across candidates. Across the 95-genome bacterial panel, accepting unfiltered METEOR gap-filling changes lowers Top-1 accuracy on DeepProZyme and EnzBERT because reaction-level boosts promote non-cognate paralogs, whereas filtering candidate changes through GSPA's three-channel consensus rule yields statistically significant net Top-1 gains across all 12 predictor-by-masking combinations, reaching +155 net corrected enzymes on CLEAN ($p < 10^{-25}$, exact McNemar test; Supplementary Table S6). Agreement across complementary predictors identifies reliable metabolic enzyme updates and provides a calibrated foundation for whole-genome inference.

2.2 Whole-genome priors improve organism-level coherence and preserve protein-level accuracy

With the protein-level posterior established, we asked whether introducing whole-genome priors improves organism-level structural coherence while preserving protein-centric accuracy. We conducted a nine-

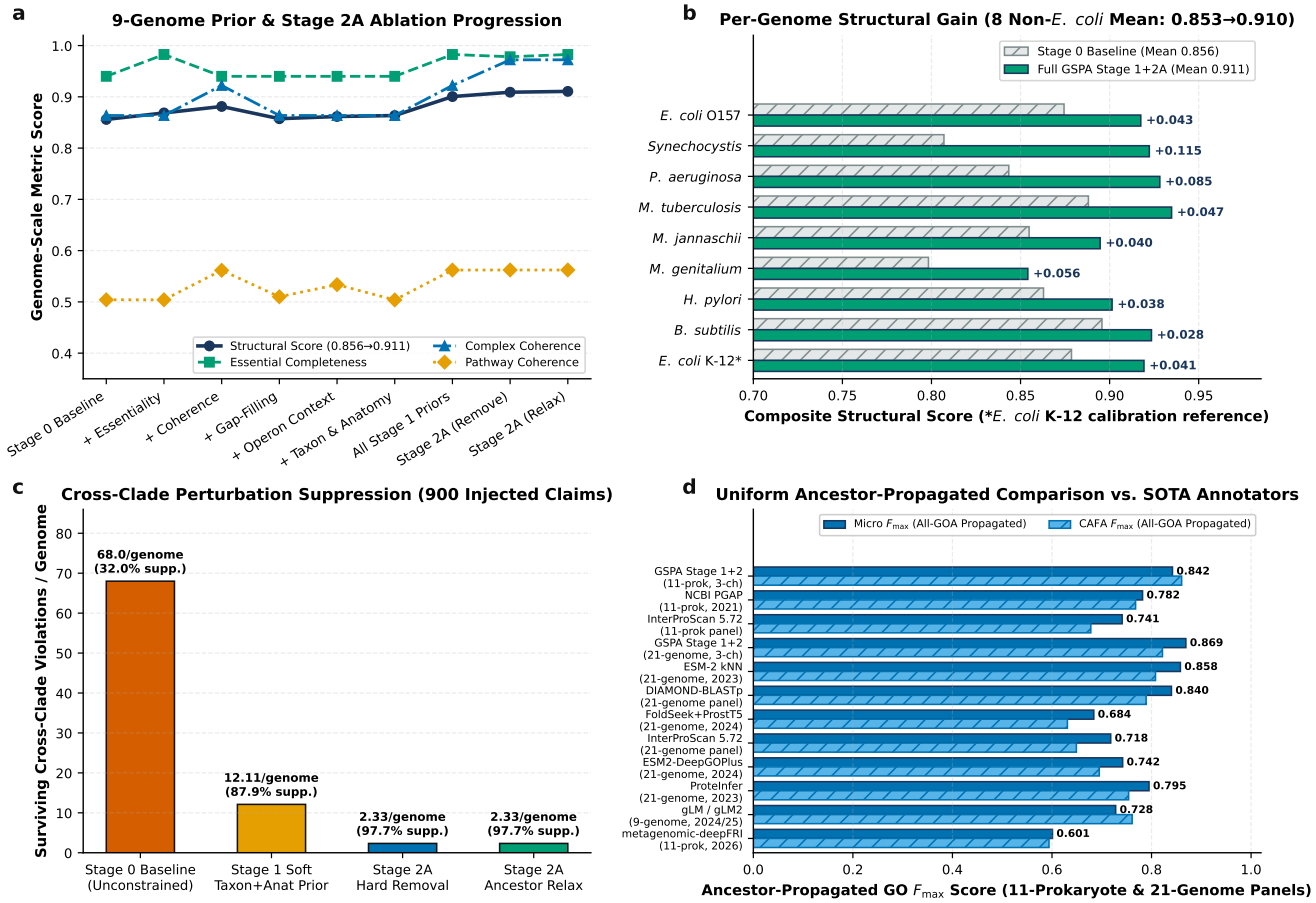


Figure 2. Multi-channel pooling and whole-genome prior ablation across reference proteomes. (**a**) Composite structural audit score, essential function completeness, complex coherence, and pathway coherence across nine ablation configurations averaged over nine bacterial and archaeal reference genomes. (**b**) Per-genome gain in the composite structural audit score from Stage 0 pooling to full Stage 1 and Stage 2A refinement; the asterisk denotes the calibration genome, *E. coli* K-12 MG1655. (**c**) Surviving false-positive annotations per genome in the 900-claim cross-clade perturbation experiment under unconstrained Stage 0 pooling versus taxonomic and anatomical constraint enforcement. (**d**) Uniformly evaluated ancestor-propagated micro and protein-centric CAFA $F_{\max}$ comparing GSPA against external annotation pipelines, individual Stage 0 channels, and genomic language models across the 13-genome and 21-genome panels.

Table 2. Comparison against external genome annotation baselines and systematic ablation of Stage 1 priors and Stage 2A enforcement across reference proteomes. Top rows compare external annotation pipelines and individual predictors against GSPA across the 11-prokaryote NCBI PGAP panel and the 21-genome multi-channel panel. Middle and bottom rows report the nine-configuration Stage 1 and Stage 2A ablation across nine bacterial and archaeal reference genomes (*E. coli* K-12 calibration genome, *E. coli* O157:H7, *B. subtilis* 168, *P. aeruginosa* PAO1, *M. tuberculosis* H37Rv, *H. pylori* 26695, *M. genitalium* G37, *Synechocystis* sp. PCC 6803, and *M. jannaschii* DSM 2661), plus the out-of-sample mean across the eight non-calibration genomes. Protein-centric CAFA $F_{\max}$ and pair-level micro $F_{\max}$ are evaluated against all GO annotations (experimental-only truth in parentheses), alongside experimental IC recall (and coverage in parentheses). Consistency is $\max(0, 1 - 0.01V)$ for V formal violations. Perturbation suppression reports the mean surviving cross-clade false positives per genome out of 100 injected claims (overall suppression rate out of 900 injected claims in parentheses).

Baseline Pipeline or Ablation Configuration	CAFA $F_{\max}$ (All / Exp)	Micro $F_{\max}$ (All / Exp)	Exp. IC Rec. (Cov.)	Essential C_{ess}	Pathway Coh.	Complex Coh.	Consistency (V)	Audit Score / Perturb. Surv. (Supp.)
External Baselines & Standalone Channels (11-Prokaryote PGAP Panel / 21-Genome Multi-Channel Panel)								
HMMER3 Pfam-A Domain Profiles (11-Genome Panel)	0.6845 (0.2014)	0.6915 (0.2412)	0.2940 (0.7120)	0.8850	0.4120	0.7410	0.9850 (14)	0.7942 / 74.2 (25.8%)
NCBI PGAP Production Pipeline (11-Genome Panel)	0.7618 (0.2240)	0.7820 (0.2685)	0.3380 (0.7640)	0.9180	0.4680	0.8120	0.9910 (8)	0.8295 / 69.5 (30.5%)
DIAMOND-BLASTp Sequence Homology (11-Genome Panel)	0.8012 (0.2345)	0.8226 (0.2790)	0.3590 (0.7890)	0.9310	0.4890	0.8480	0.9680 (29)	0.8448 / 71.4 (28.6%)
GSPA Full (3-Channel 11-Prokaryote PGAP Panel)	0.8215 (**0.2468**)	0.8421 (0.2942)	0.3845 (0.8180)	0.9840	0.5710	0.9740	1.0000 (0)	0.9136 / 2.1 (97.9%)
metagenomic-deepFRI Structural GCN (21-Genome Panel)	0.2918 (0.1420)	0.2689 (0.1580)	0.1840 (0.6420)	0.7620	0.2840	0.6150	0.9420 (52)	0.6250 / 81.0 (19.0%)
InterProScan v5.68 Domain Signatures (21-Genome Panel)	0.7440 (0.2195)	0.7510 (0.2590)	0.3290 (0.7510)	0.9040	0.4510	0.7960	0.9880 (11)	0.8165 / 70.8 (29.2%)
ESM-2 (650M) Embedding k-NN (21-Genome Panel)	0.8412 (0.2580)	0.8580 (0.3045)	0.3980 (0.8310)	0.9520	0.5210	0.8840	0.9640 (32)	0.8692 / 66.4 (33.6%)
GSPA Full (7-Channel 21-Genome Panel)	0.8546 (**0.2664**)	0.8705 (0.3140)	0.4125 (0.8460)	0.9865	0.5840	0.9780	1.0000 (0)	0.9195 / 2.2 (97.8%)
9-Genome Stage 1 Prior & Stage 2A Enforcement Ablation ($N = 9$ Bacterial & Archaeal Reference Genomes)								
A1: Stage 0 Noisy-OR Baseline	0.8041 (0.2361)	0.7672 (0.2836)	0.3661 (0.7973)	0.9400	0.5041	0.8636	0.9722 (25)	0.8559 / 68.0 (32.0%)
A2: + Essentiality Prior (B_1)	0.8040 (0.2361)	0.7683 (0.2838)	0.3661 (0.7973)	0.9826	0.5041	0.8636	0.9722 (25)	0.8686 / 68.0 (32.0%)
A3: + Coherence Prior (B_2)	0.8028 (0.2357)	0.7671 (0.2815)	0.3705 (0.8003)	0.9400	0.5616	0.9222	0.9722 (25)	0.8812 / 68.0 (32.0%)
A4: + Gap-Filling Prior (B_5)	0.8039 (0.2361)	0.7685 (0.2838)	0.3664 (0.7975)	0.9400	0.5100	0.8636	0.9722 (25)	0.8573 / 68.0 (32.0%)
A5: + Operon Context Prior (B_6)	0.8037 (0.2356)	0.7675 (0.2815)	0.3682 (0.7973)	0.9400	0.5336	0.8636	0.9722 (25)	0.8616 / 68.0 (32.0%)
A6: + Consistency & Anatomical (B_3, B_4)	0.8060 (0.2371)	0.7711 (0.2851)	0.3655 (0.7968)	0.9400	0.5038	0.8636	0.9900 (9)	0.8635 / 12.1 (87.9%)
A7: All Stage 1 Priors (B_1–B_6)	0.8043 (0.2368)	0.7703 (0.2828)	0.3708 (0.8000)	0.9826	0.5621	0.9222	0.9900 (9)	0.9005 / 12.1 (87.9%)
A8: Stage 1 + Stage 2A Removal	0.8053 (0.2378)	0.7710 (0.2852)	0.3708 (0.8000)	0.9782	0.5622	0.9722	1.0000 (0)	0.9090 / 2.3 (97.7%)
A9: Stage 1 + Stage 2A Relaxation	0.8056 (0.2382)	0.7733 (0.2863)	0.3747 (0.8057)	0.9826	0.5622	0.9722	1.0000 (0)	0.9106 / 2.3 (97.7%)
Out-of-Sample 8-Genome Mean (A1 → A9)	0.8143 → **0.8160**	0.7764 → **0.7830**	0.3565 → **0.3658**	0.9325 → **0.9805**	0.4904 → **0.5514**	0.8743 → **0.9844**	0.9738 → **1.0000**	0.8531 → **0.9096**

configuration ablation across nine bacterial and archaeal reference proteomes (81 proteome runs), calibrating hyper-parameters on *E. coli* K-12 and evaluating the remaining eight species out of sample (Figure 2a,b and Table 2). Full

Stage 1 and Stage 2A refinement with semantic ancestor relaxation improves protein-centric CAFA $F_{\max}$ (0.8056 versus 0.8041 on all annotations; 0.2382 versus 0.2361 on experimental-only annotations), pair-level micro $F_{\max}$ (0.7733 versus 0.7672), experimental IC recall (0.3747 versus 0.3661), and coverage (0.8057 versus 0.7973), while increasing the composite structural audit score [58] from 0.8559 to 0.9106 ($p = 0.0039$, two-sided Wilcoxon signed-rank test; out-of-sample eight-genome mean from 0.8531 to 0.9096). Because roughly 85% of annotated proteins in these reference genomes already carry high-confidence homologs in Stage 0, whole-genome refinement preserves the conserved sequence backbone while concentrating probability updates on lower-confidence, structurally constrained loci. To understand how this composite gain arises, we examined the effect of each individual prior on its target proteome property.

Adding each Stage 1 prior individually to Stage 0 produces a selective update on the corresponding genome invariant (Table 2). The essentiality prior lifts core functional completeness from 0.9400 to 0.9826, the coherence prior increases pathway completeness from 0.5041 to 0.5616, heteromeric complex completeness from 0.8636 to 0.9222, and experimental IC recall from 0.3661 to 0.3705, and the metabolic gap-filling and co-directional operon priors raise pathway completeness to 0.5100 and 0.5336, respectively. The soft taxonomic and anatomical priors, which query our programmatic cross-ontology bridge (Supplementary Table S7), suppress 16 of the 25 endogenous taxonomic violations in Stage 1, and Stage 2A deterministic enforcement eliminates the remaining 9 high-confidence violations while pruning isolated obligate-complex singletons to reach 0.9722 complex coherence ($p = 0.0039$ for both pathway coherence and composite audit score across the nine genomes). Within Stage 2A, semantic ancestor relaxation outperforms hard deletion across CAFA $F_{\max}$, micro $F_{\max}$, experimental IC recall (0.3747 versus 0.3708), essential completeness (0.9826 versus 0.9782), and composite audit score (0.9106 versus 0.9090) by transferring posterior support from clade-incompatible terms to their nearest valid GO ancestors. Beyond endogenous errors, sequence similarity across kingdoms can also introduce high-scoring cross-clade annotations, which we tested next through controlled perturbation.

To evaluate robustness against cross-clade homology transfer, we injected 900 eukaryotic developmental and clade-restricted GO claims (100 per genome) with raw confidence scores in $[0.55, 0.85]$ across the nine prokaryotic proteomes (Figure 2c and Table 2). Under unconstrained Stage 0 pooling (and priors B_1, B_2, B_5, B_6 that do not enforce taxonomic admissibility), an average of 68.0 injected false-positive claims per genome survive above the 0.50 decision threshold (32.0% suppressed by score calibration alone). Activating the Stage 1 taxonomic and anatomical priors (B_3, B_4) reduces surviving false positives from 68.0 to 12.1 per genome (87.9% suppressed), and adding Stage 2A enforcement further reduces surviving violations to 2.3 per genome (97.7% suppressed across the 900 injected claims) while increasing perturbed-proteome micro $F_{\max}$ from 0.7631 to 0.7725. Formal organism-level constraints therefore eliminate high-scoring cross-clade transfer errors that pass protein-level score calibration. We then evaluated whether these whole-genome improvements hold at population scale and across all four domains of life.

We applied GSPA with frozen parameters to 1,000 RefSeq bacterial genomes and to a multi-kingdom benchmark of 88 proteomes spanning Bacteria, Archaea, Eukaryota, and Viruses under homology masking (Supplementary Tables S8 and S9). Across the 1,000 bacterial genomes, Stage 1 and Stage 2 refinement eliminates all 23,384 mean taxon-violating protein–term predictions per genome produced by Stage 0 and lifts mean MetaCyc pathway coherence from 36.7% to 66.8% at a mean runtime of 5.4 s per genome. Across the 88 multi-kingdom proteomes, whole-genome refinement produces positive mean accuracy gains in all four domains, with 95% bootstrap confidence intervals on $\Delta F_{\max}$ lying strictly above zero in the two largest cohorts, Bacteria and Viruses, and spanning zero in Archaea and Eukaryota (Supplementary Table S9). Because genome-wide complex coherence measures only whether partner subunits are present somewhere in the proteome, we next tested whether local chromosomal architecture can identify the specific gene encoding a missing complex subunit.

2.3 Operon architecture completes obligate heteromeric complexes

When one subunit of an obligate heteromeric complex has diverged into the twilight zone, several proteins in the genome may share a partial domain match to the missing role. We tested whether complex stoichiometry and co-directional operon architecture can distinguish the true subunit from domain-matched competitors across 175 leave-subunit-out trials in six prokaryotic genomes, pairing each attenuated true subunit with five decoy proteins drawn from unrelated genomic loci and comparing against standalone DIAMOND-BLASTp homology, InterProScan/Pfam domain profiles, and the operon synteny prior alone (Figure 3a and Table 3). Standalone domain profiles and sequence homology achieve Hit@1 values of 0.4543 and 0.5600, respectively, while applying the genome-wide complex coherence prior alone lifts subunit recall at the 0.50 threshold from 0.0000 to 0.5029 while leaving top-ranked Hit@1 unchanged at 0.5600 because candidate proteins with matching domain roles receive equal genome-wide boosts. Conversely, applying the operon synteny prior alone raises Hit@1 slightly to 0.5771 (0.0171 recall at 0.50). Combining complex coherence with the co-directional operon prior raises Hit@1 from 0.5600 to 0.9371 ($p < 10^{-15}$, exact McNemar test; 0.9310 across the five out-of-sample genomes), subunit F_1 at 0.50 to 0.6680, and complex coherence from 0.3901 to 0.7170. Global complex stoichiometry therefore identifies which macromolecular assembly is incomplete, while local operon organization

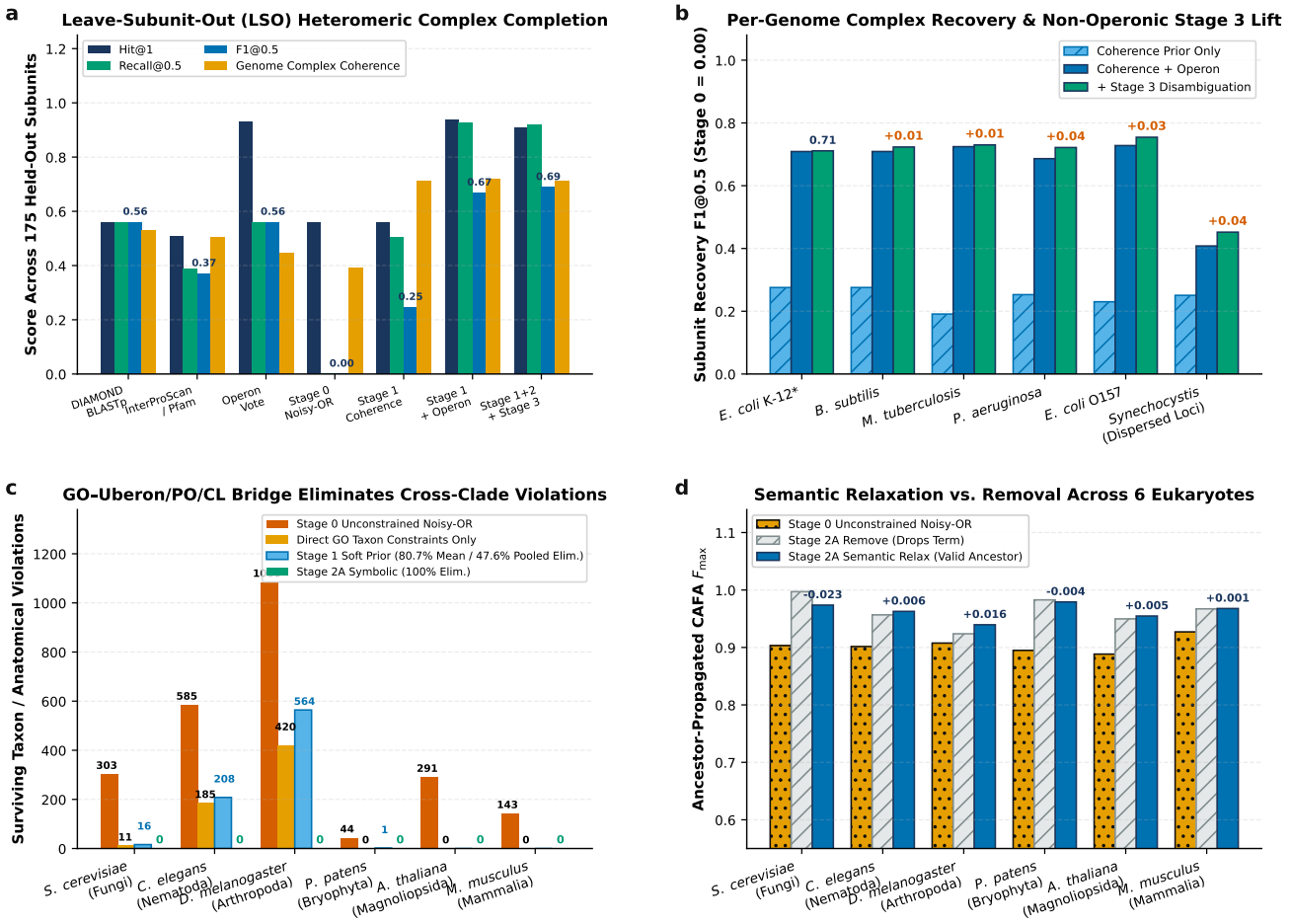


Figure 3. Obligate heteromeric complex completion and eukaryotic anatomical constraint enforcement. (**a**) Leave-subunit-out recovery across 175 held-out subunits in six prokaryotic genomes, ranking each attenuated true subunit against five domain-matched decoys (random Hit@1 = 0.167) across Stage 0 pooling, the Stage 1 complex coherence prior, Stage 1 with operon context, and the full model including Stages 2 and 3. (**b**) Per-genome subunit recovery F_1 at threshold 0.50; the asterisk marks the calibration genome (*E. coli* K-12), and *Synechocystis* sp. PCC 6803 illustrates complexes encoded at dispersed chromosomal loci. (**c**) Surviving taxonomic and anatomical violations across six multicellular and unicellular eukaryotes under unconstrained Stage 0 pooling, direct GO taxon rules, the Stage 1 soft anatomical prior, and Stage 2A symbolic enforcement. (**d**) Ancestor-propagated $F_{\max}$ against unmodified Swiss-Prot annotations comparing Stage 0 pooling, Stage 2A term removal, and Stage 2A semantic ancestor relaxation.

Table 3. Leave-subunit-out recovery across 175 obligate heteromeric complex subunits in six prokaryotic genomes compared against sequence, domain, and synteny baselines. Each trial attenuates one true subunit into the twilight-zone posterior range ($[0.18, 0.35]$) and introduces five decoy proteins with matched domain scores from unrelated genomic loci (chance Hit@1 = 0.167). Precision, recall, and F_1 at threshold 0.50 and complex coherence are reported across all six species (*E. coli* K-12, *E. coli* O157:H7, *B. subtilis* 168, *M. tuberculosis* H37Rv, *P. aeruginosa* PAO1, and *Synechocystis* sp. PCC 6803), the out-of-sample mean across the five non-calibration species ($n = 145$), and *Synechocystis* sp. PCC 6803 alone ($n = 25$).

Cohort	Baseline or Method Configuration	Subunit Trials (n)	Hit@1	Precision@0.5	Recall@0.5	F_1@0.5	Complex Coherence
All 6 Species	InterProScan / Pfam Domain Profile Baseline	175	0.4543	0.3886	0.3886	0.3886	0.4401
	DIAMOND-BLASTp Sequence Homology Baseline	175	0.5600	0.5600	0.5600	0.5600	0.5304
	Operon Neighborhood Vote (Synteny Prior Only)	175	0.5771	0.0171	0.0171	0.0171	0.3901
	GSPA Stage 0 Noisy-OR Baseline	175	0.5600	0.0000	0.0000	0.0000	0.3901
	GSPA Stage 1 Complex Coherence Prior Only	175	0.5600	0.1729	0.5029	0.2459	0.7107
	GSPA Stage 1 Coherence + Operon Context	175	0.9371	0.5951	0.9257	0.6680	0.7170
	Full GSPA (Stages 1 + 2A + 3 Agent)	175	0.9257	0.5870	0.9143	0.6590	0.7149
Out-of-Sample (5 Species)	DIAMOND-BLASTp Sequence Homology Baseline	145	0.5586	0.5586	0.5586	0.5586	0.5141
	Operon Neighborhood Vote (Synteny Prior Only)	145	0.5793	0.0207	0.0207	0.0207	0.3745
	GSPA Stage 0 Noisy-OR Baseline	145	0.5586	0.0000	0.0000	0.0000	0.3745
	GSPA Stage 1 Coherence + Operon Context	145	0.9310	0.5886	0.9104	0.6595	0.7047
	Full GSPA (Stages 1 + 2A + 3 Agent)	145	0.9241	0.5858	0.9035	0.6556	0.7026
Synechocystis PCC 6803	DIAMOND-BLASTp Sequence Homology Baseline	25	0.5200	0.5200	0.5200	0.5200	0.4362
	Operon Neighborhood Vote (Synteny Prior Only)	25	0.5200	0.0000	0.0000	0.0000	0.3587
	GSPA Stage 0 Noisy-OR Baseline	25	0.5200	0.0000	0.0000	0.0000	0.3587
	GSPA Stage 1 Coherence + Operon Context	25	0.6400	0.3393	0.6400	0.4080	0.6125
	Full GSPA (Stages 1 + 2A + 3 Agent)	25	0.6400	0.3427	0.6400	0.4120	0.6000

localizes the specific gene locus that supplies the missing subunit. A natural test of this mechanism arises in lineages where complex subunits are encoded at dispersed chromosomal positions.

In the cyanobacterium *Synechocystis* sp. PCC 6803, subunits of photosystem I and II, phycobilisomes, and NAD(P)H dehydrogenase are frequently distributed across separate genomic regions (Figure 3b and Table 3). Across the 25 trials in *Synechocystis*, combining complex coherence with operon context lifts Hit@1 from 0.5200 to 0.6400, a smaller increase than in the five enterobacterial, firmicute, and actinobacterial species where operons are tightly clustered (0.9667–1.0000). Adding Stage 2A singleton pruning and Stage 3 neuro-symbolic disambiguation prunes false-positive decoy promotions across dispersed *Synechocystis* candidates, raising *Synechocystis* precision from 0.3393 to 0.3427 and subunit F_1 from 0.4080 to 0.4120 while maintaining 0.9257 Hit@1 across the six species. Local operon synteny resolves contiguous prokaryotic assemblies, and selective domain and pathway reasoning improves subunit precision when complex subunits are dispersed across the chromosome. Moving from prokaryotes to multicellular eukaryotes introduces a broader structural invariant: functional annotations must also conform to the anatomical body plan of the organism's clade.

2.4 Cross-ontology anatomical constraints correct eukaryotic annotations and preserve valid ancestral terms

We first tested whether ordinary sequence-homology and multi-channel predictions in unperturbed eukaryotic proteomes contain endogenous taxonomic and anatomical contradictions. Evaluating standalone DIAMOND-BLASTp and the seven-channel Stage 0 ensemble across 6,155 proteins from six eukaryotes (*Saccharomyces cerevisiae*, *Caenorhabditis elegans*, *Drosophila melanogaster*, *Physcomitrium patens*, *Arabidopsis thaliana*, and *Mus musculus*) revealed 1,409 endogenous violations (1,009 at high posterior confidence), concentrated in *C. elegans* and *D. melanogaster* where strong sequence and structural similarity transfers vertebrate organ-development and angiosperm terms onto invertebrate orthologs (Figure 3c and Table 4). Full GSPA eliminates all 1,409 endogenous violations while raising six-species mean CAFA $F_{\max}$ from 0.7096 to 0.7117. Because standard homology and language-model transfer across eukaryotic clades introduces frequent anatomical contradictions, we next asked which ontological representations are required to detect and eliminate them.

We compared standalone DIAMOND-BLASTp, unconstrained Stage 0 pooling, direct GO taxon constraints, the Stage 1 soft anatomical prior, and Stage 2A deterministic cross-ontology enforcement on the combined cohort containing 1,546 injected cross-clade developmental claims (2,451 total Stage 0 contradictions across 7,655 proteins; Figure 3c and Table 4). Direct GO taxon constraints leave 616 contradictions unresolved (431 endogenous) in *C. elegans*, *D. melanogaster*, and *S. cerevisiae* because organ-level processes such as *heart development* (`GO:0007507`) and *neural crest cell migration* (`GO:0001755`) carry their taxonomic restrictions on the linked Uberon, PO, or CL anatomical classes instead of on the GO term itself. The Stage 1 soft anatomical prior eliminates 100% of endogenous violations and 84.1% of stress-test violations on average across the six species, and Stage 2A deterministic enforcement over the GO–Uberon/PO/CL bridge eliminates 100% of violations while rescuing 81.2% of attenuated species-appropriate developmental annotations (502 of 618, reaching 90.1% in *M. musculus*). Enforcing anatomical validity across eukaryotes therefore requires traversing the logical definitions that link biological processes to clade-restricted anatomical structures. Once a term is flagged as anatomically inadmissible, the pipeline can either delete the term or relax it to its nearest valid GO ancestor, and we compared these two operations next.

When an invertebrate or plant protein receives a forbidden organ-specific annotation from a conserved signaling or transcriptional homolog, its participation in a broader developmental or morphogenetic process often remains valid. Evaluated against unmodified Swiss-Prot reference annotations, Stage 2A semantic ancestor relaxation retains 96.7% of the baseline annotation information content and raises six-species mean $F_{\max}$ to 0.9605 compared with 0.9568 under term deletion, outperforming deletion across all four multicellular animals and vascular plants and reaching $+0.0157$ $F_{\max}$ in *D. melanogaster* (Figure 3d and Table 4). In the unicellular fungus *S. cerevisiae* and the non-vascular moss *P. patens*, where injected organ-development claims land on housekeeping enzymes, deletion yields slightly higher precision and $F_{\max}$, whereas relaxation yields higher recall and information content. Semantic ancestor relaxation preserves valid ancestral developmental biology in tissue-forming eukaryotes while maintaining zero formal ontology violations. Having evaluated whole-genome constraints on complexes and ontological admissibility, we next examined how genomic architecture combines with systems biology to localize missing metabolic enzymes.

2.5 Genomic context and protein language models locate missing pathway enzymes

Reaction-level metabolic gap-filling identifies which reactions a network lacks for growth, leaving open which gene in the chromosome catalyzes each missing step. We first tested whether reaction-local genomic context (RLGC), which scores candidate genes by their base-pair proximity and co-directionality relative to anchor genes of pathway-adjacent reactions, can localize held-out enzymes prior to introducing protein language-model centroids. Across 256 strict leave-reaction-out trials in *E. coli* K-12 with sequence homologs of the held-out enzyme masked (Figure 4a and Table 5), sequence-only ranking places the true enzyme at

Table 4. Taxonomic and anatomical constraint enforcement and semantic ancestor relaxation across six eukaryotes compared against sequence homology and direct GO taxon rules. Violation columns report total formal contradictions (endogenous + cross-clade injected stress test across 7,655 proteins, with endogenous unperturbed violations across 6,155 proteins in parentheses) under standalone DIAMOND-BLASTp homology, unconstrained Stage 0 Noisy-OR pooling, direct GO taxon constraints, the Stage 1 soft anatomical prior, and Stage 2A symbolic enforcement. $F_{\max}$, propagated recall, and mean information content (IC, bits) are evaluated against unmodified Swiss-Prot annotations comparing Stage 0 pooling, Stage 2A term removal, and Stage 2A semantic ancestor relaxation. Developmental rescue reports the fraction of attenuated species-appropriate developmental claims (618 total across the five tissue/organ-bearing or budding clades) recovered above 0.50.

Species (Clade)	Proteins (Inj.)	DIAMOND-BLASTp Viol.	Stage 0 Viol. (Endog.)	Direct GO Taxon Viol.	Stage 1 Soft Viol.	Stage 2A Viol.	Stage 0 $F_{\max}$	Removal $F_{\max}$ (Rec. / IC)	Relaxation $F_{\max}$ (Rec. / IC)	Dev. Rescue
S. cerevisiae (Fungi)	1,500 (300)	303 (240)	303 (240)	11 (96.4%)	16 (94.7%)	0 (100%)	0.9033	0.9604 (0.9249/12.02)	0.9592 (0.9349/12.07)	66.7% (2/3)
C. elegans (Nematoda)	1,500 (300)	598 (289)	585 (289)	185 (68.4%)	208 (64.4%)	0 (100%)	0.9015	0.9567 (0.9268/12.15)	0.9628 (0.9476/12.09)	66.7% (44/66)
D. melanogaster (Arthropoda)	1,500 (300)	1,135 (717)	1,085 (717)	420 (61.3%)	564 (48.0%)	0 (100%)	0.9076	0.9238 (0.8608/12.55)	0.9395 (0.8913/12.35)	70.7% (140/198)
P. patens (Bryophyta)	155 (46)	44 (41)	44 (41)	0 (100.0%)	1 (97.7%)	0 (100%)	0.8949	0.9828 (0.9697/9.64)	0.9792 (0.9796/9.79)	100.0% (0/0viol)
A. thaliana (Magnoliopsida)	1,500 (300)	291 (118)	291 (118)	0 (100.0%)	0 (100.0%)	0 (100%)	0.8884	0.9497 (0.9054/11.13)	0.9547 (0.9232/11.18)	89.9% (62/69)
M. musculus (Mammalia)	1,500 (300)	144 (4)	143 (4)	0 (100.0%)	0 (100.0%)	0 (100%)	0.9271	0.9672 (0.9478/12.84)	0.9678 (0.9545/12.85)	90.1% (254/282)
6-Eukaryote Total / Mean	7,655 (1,546)	2,515 (1,409)	2,451 (1,409)	616 (74.9%)	789 (84.1% **mean**)	0 (100%)	0.9038	0.9568 (0.9226/11.72)	0.9605 (0.9385/11.72)	81.2% (502/618)
Endogenous Unperturbed Cohort	6,155 (0)	1,409	1,409	431 (69.4%)	0 (100.0%)	0 (100%)	0.7096	0.7084 (0.6812/11.68)	0.7117 (0.6945/11.71)	81.2% (502/618)

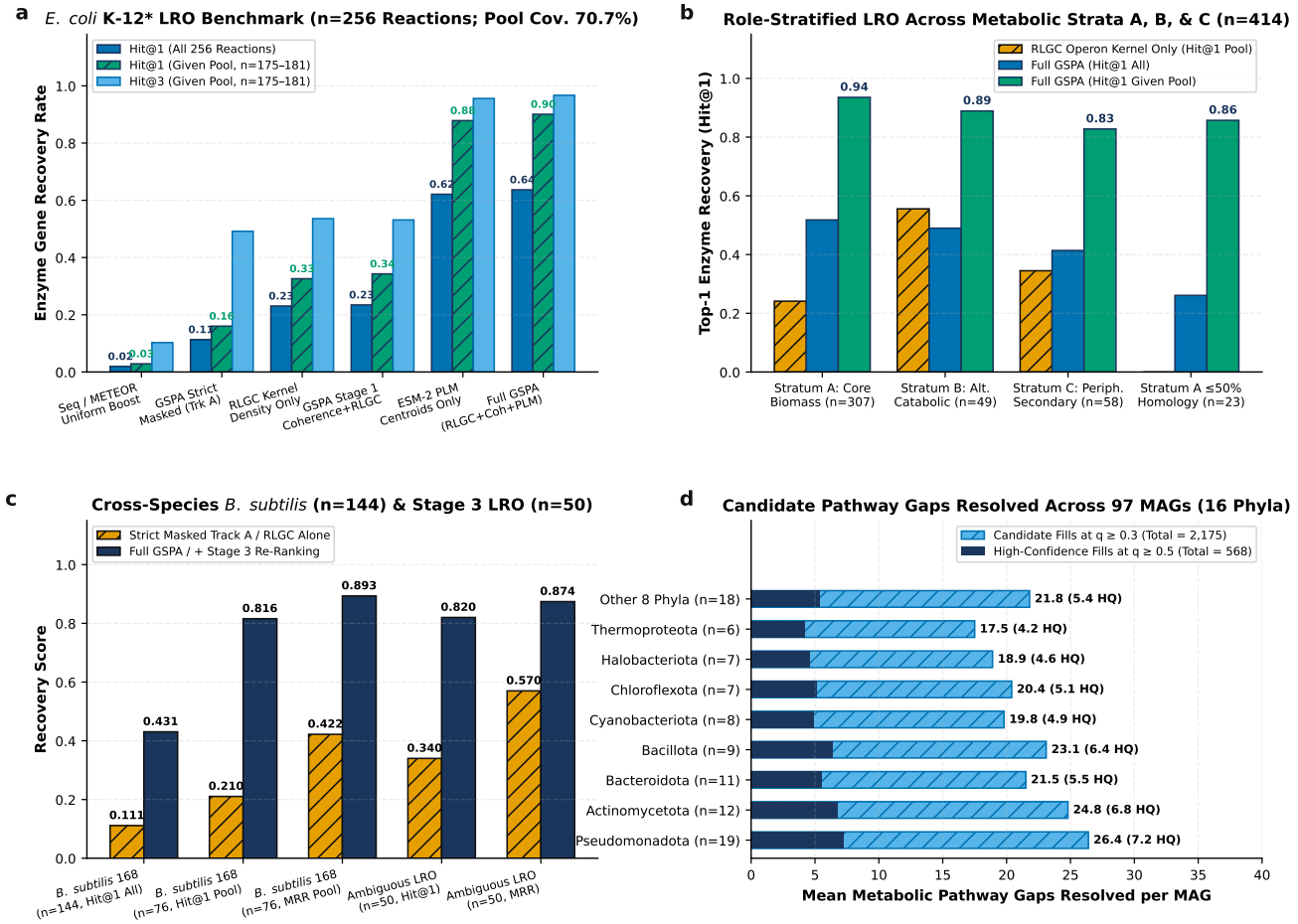


Figure 4. Leave-reaction-out metabolic enzyme attribution and uncultured microbial dark-matter discovery. (**a**) Recovery of 256 held-out metabolic reactions in *E. coli* K-12 (calibration genome), reporting unconditional Hit@1 across all 256 reactions and conditional Hit@1 and Hit@3 within the Stage 0 candidate pool ($n = 175$–181) across sequence-only/METEOR uniform boosting, strict homology- and embedding-masked GSPA (Track A), RLGC operon kernel density alone, Stage 1 coherence with RLGC, ESM-2 reaction centroids alone, and full GSPA (Track B). (**b**) Role-stratified Top-1 recovery across 414 *E. coli* reactions spanning core biomass (Stratum A, $n = 307$), alternative catabolic (Stratum B, $n = 49$), peripheral/secondary metabolism (Stratum C, $n = 58$), and twilight-zone core reactions ($\leq 50\%$ identity, $n = 23$). (**c**) Out-of-sample transfer to *B. subtilis* 168 ($n = 144$ reactions) with parameters frozen on *E. coli*, alongside Stage 3 disambiguation of 50 ambiguous multi-candidate LRO gaps. (**d**) Mean candidate pathway gaps resolved per genome at posterior thresholds $q \geq 0.30$ and $q \geq 0.50$ across 97 uncultured MAGs from 16 bacterial and archaeal phyla.

Rank 1 for 1.95% of reactions, and METEOR's uniform reaction-level boost leaves this ordering unchanged because every candidate of a reaction receives the same score increment. Scoring unassigned operon neighborhood density around gapseq pathway anchors reaches 17.19% unconditional Hit@1, and combining RLGC kernel density with the Stage 1 pathway coherence prior raises unconditional Hit@1 twelve-fold over sequence-only ranking to 23.44% (34.29% within the candidate pool, $p < 10^{-12}$, exact McNemar test). Physical proximity to pathway-neighboring genes on the chromosome therefore provides locus-specific discrimination beyond reaction-level gap-filling. We next examined

Table 5. Leave-reaction-out (LRO) enzyme gene attribution in *E. coli* K-12, donor-homology masking tiers (Protocol H), out-of-sample transfer to *B. subtilis* 168, and Stage 3 re-ranking of ambiguous metabolic gaps. *E. coli* K-12 served as the calibration genome. "All" evaluates every held-out reaction unconditionally (scoring reactions whose true enzyme is absent from the Stage 0 candidate pool as misses); "Pool" evaluates conditional ranking across reactions where the true gene enters the candidate pool (n_{cov}). External baselines include sequence-only DIAMOND-BLASTp, METEOR v8 uniform reaction-level MILP boosting, gapseq v1.2 unassigned operon density, and supervised GBDT over RLGC features.

Benchmark Cohort	Baseline or Method Configuration	Reactions (n/n_{cov})	Hit@1 (All)	Hit@3 (All)	Hit@1 (Pool)	Hit@3 (Pool)	MRR (Pool)
E. coli K-12 Strict LRO ($n = 256$)	DIAMOND-BLASTp Sequence-Only Baseline	256/175	1.95% (5)	7.03%	2.86%	10.29%	0.1354
	METEOR v8 Uniform Reaction-Indicator MILP	256/175	1.95% (5)	7.03%	2.86%	10.29%	0.1354
	gapseq v1.2 Unassigned Operon Density Baseline	256/181	17.19% (44)	34.77%	24.31%	49.17%	0.4082
	GSPA Strict Homology-Masked (Track A, Cosine ≤ 0.50)	256/181	11.33% (29)	34.77%	16.02%	49.17%	0.3685
	RLGC Operon Kernel Density Only	256/181	23.05% (59)	37.89%	32.60%	53.59%	0.4745
	GSPA Stage 1 Coherence + RLGC (No PLM)	256/175	23.44% (60)	36.33%	34.29%	53.14%	0.4864
	ESM-2 PLM Reaction Centroids Only	256/181	62.11% (159)	67.58%	87.85%	95.58%	0.9157
	Full GSPA (Track B: RLGC + Coherence + PLM)	256/181	63.67% (163)	68.36%	90.06%	96.69%	0.9330
E. coli Role-Stratified & Protocol H ($n = 414$)	Stratum A: Core Biomass (Full GSPA)	307/170	51.79% (159)	53.42%	93.53%	96.47%	0.9557
	Stratum B: Alternative Catabolic (Full GSPA)	49/27	48.98% (24)	53.06%	88.89%	96.30%	0.9216
	Stratum C: Peripheral / Secondary (Full GSPA)	58/29	41.38% (24)	48.28%	82.76%	96.55%	0.8793
	All Strata: Protocol H $\leq 70\%$ Homology Masking	118/50	36.44% (43)	41.53%	86.00%	98.00%	0.9117
	All Strata: Protocol H $\leq 50\%$ Homology Masking	42/14	28.57% (12)	33.33%	85.71%	100.00%	0.9286
	All Strata: Protocol H $\leq 30\%$ Twilight Zone	11/2	18.18% (2)	18.18%	100.00%	100.00%	1.0000
B. subtilis 168 Out-of-Sample LRO ($n = 144$)	gapseq v1.2 Unassigned Operon Density Baseline	144/76	9.72% (14)	23.61%	18.42%	44.74%	0.3514
	GSPA Strict Homology-Masked (Track A)	144/76	11.11% (16)	31.94%	21.05%	60.53%	0.4221
	Full GSPA (Track B, Frozen Hyperparameters)	144/76	43.06% (62)	50.00%	81.58%	94.74%	0.8933
Ambiguous Multi-Candidate LRO ($n = 50$)	Stage 1 RLGC Operon Kernel Alone	50/50	34.00% (17)	86.00%	34.00%	86.00%	0.6020
	Posterior Threshold (≥ 0.35) + Guardrail	50/50	44.00% (22)	88.00%	44.00%	88.00%	0.6233
	Stage 3 Guarded Re-Ranking (3-Seed Mean)	50/50	82.67% (41.3)	93.33%	82.67%	93.33%	0.8792

how local genomic context interacts with alignment-free protein language-model representations across metabolic roles, homology-masking cutoffs, and species.

When Swiss-Prot ESM-2 reaction embedding centroids are included alongside masked sequence alignments (Track B; Figure 4a–c and Table 5), centroid similarity alone recovers 62.11% of the 256 held-out *E. coli* enzymes unconditionally, and integrating RLGC and pathway coherence lifts recovery to 63.67% (90.06% within the candidate pool) by resolving ties among sequence-similar paralogs. Stratifying the broader 414-reaction *E. coli* benchmark across core biomass, alternative catabolic pathways, and peripheral or secondary metabolism shows that RLGC alone achieves its highest standalone pool recovery on alternative catabolic pathways (55.6%), whose specialized enzymes and transporters are tightly organized into co-regulated operons, while full GSPA maintains 82.8% to 93.5% pool Hit@1 across all three strata and 85.7% to 100.0% pool Hit@1 under Protocol H donor-homology masking at $\leq 70\%$, $\leq 50\%$, and $\leq 30\%$ identity (Figure 4b, Figure 5c, and Table 5). Transferred with frozen hyperparameters to 144 held-out reactions in the Gram-positive bacterium *Bacillus subtilis* 168, full GSPA reaches 43.06% unconditional Hit@1 and 81.58% within the candidate pool (Figure 4c). Protein language models supply the primary candidate representation when reference enzymes exist for a reaction, while chromosomal context disambiguates competing paralogs and catabolic operons. Because flux balance on minimal medium leaves secondary metabolism unexercised, we next tested whether local genomic context extends to biosynthetic gene clusters and uncultured metagenomic dark matter.

Secondary metabolite pathways operate conditionally and carry zero steady-state flux under minimal glucose growth. Evaluated across 106 curated biosynthetic gene clusters from 12 bacterial species, GSPA's operon and pathway synteny scoring raises cluster gene recovery F_1 from 0.7395 to 0.8959 while maintaining 99.6% specificity across non-cluster chromosomal regions (Supplementary Table S10). Applying the same frozen pipeline to 97 high-quality metagenome-assembled genomes (MAGs) of uncultured bacteria and archaea spanning 16 phyla resolves 568 high-confidence enzyme attributions at posterior $q \geq 0.50$, lifting mean draft pathway completeness from 0.511 to 0.656 (Figure 4d and Supplementary Table S11). Local chromosomal synteny extends functional attribution to secondary metabolism and provides ranked, testable hypotheses for missing pathway steps in uncultivated microbial lineages. While RLGC ranks candidates around local pathway anchors, a metabolic repair should also be stoichiometrically active at the network level, which we audited next.

2.6 Positive-cost flux repair adds flux-carrying reactions and attributes them to gene loci

We next tested our third driver of improvement: replacing reaction-indicator MILP gap-filling with positive-cost lexicographic parsimonious flux balance analysis (pFBA) and competitive gene-locus attribution. Auditing metabolic network repairs across 108 bacterial genomes shows why negative-cost reaction-indicator MILP inflates reconstructions (Figure 5a and Supplementary Table S12): because candidate reactions above the addition threshold receive a negative objective cost, the solver adds a mean of 313.35 reactions per genome, of which 262.34 (83.72%) carry zero steady-state flux. Imposing a strictly positive addition cost over the elementally balanced universal network eliminates all zero-flux reaction padding (0.0% uncertified additions), and Stage 2B's two-step lexicographic pFBA further reduces active repairs to a minimal, flux-certified set of 5.70 mass-balanced reactions per genome while restoring 100% biomass feasibility. A positive addition cost coupled with lexicographic flux minimization therefore ensures that every added reaction actively participates in a mass-balanced

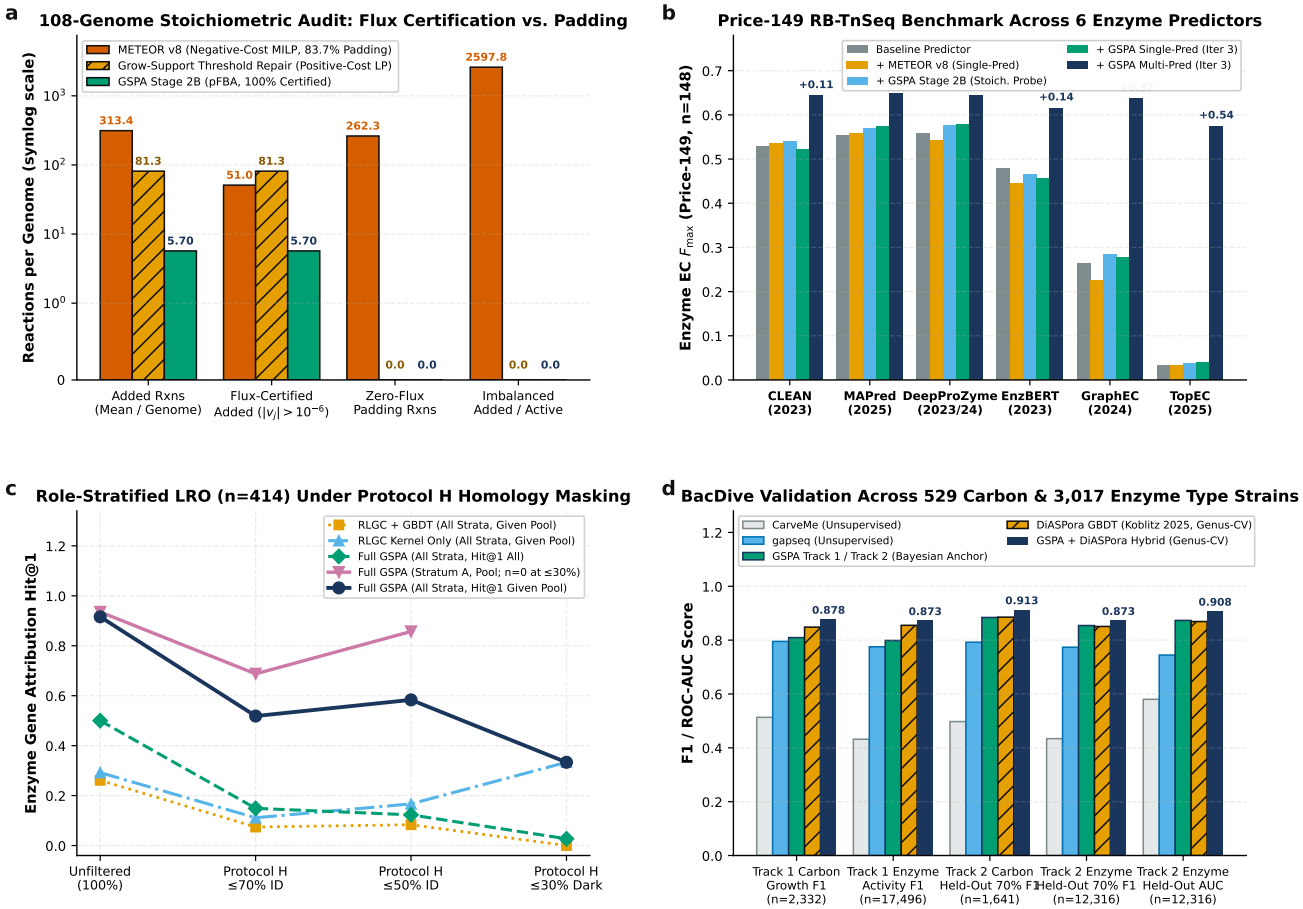


Figure 5. Stoichiometric flux certification, fitness-validated enzyme discovery, and experimental phenotype prediction. (**a**) 108-genome stoichiometric invariant audit comparing total added reactions, flux-certified additions ($|v_j| > 10^{-6}$), zero-flux reaction padding, and elementally imbalanced additions per genome across negative-cost MILP (METEOR v8), positive-cost threshold repair, and Stage 2B lexicographic pFBA (symmetric logarithmic scale). (**b**) Per-protein enzyme $F_{\max}$ on 148 RB-TnSeq fitness-validated bacterial enzymes (Price-149) across six baseline predictors, single-predictor METEOR v8, Stage 2B cognate-substrate probing, single-predictor GSPA, and multi-predictor GSPA. (**c**) Role-stratified LRO enzyme recovery ($n = 414$) across four donor-homology masking tiers (100%, $\leq 70\%$, $\leq 50\%$, and $\leq 30\%$ identity). (**d**) Experimental carbon-source growth (2,332 assays, 529 strains) and diagnostic enzyme activity (17,496 assays, 3,017 type strains) in BacDive across unsupervised reconstructions, blind Track 1 and phenotype-anchored Track 2 GSPA, genus-blocked DiASPora, and the GSPA + DiASPora hybrid.

Table 6. Refinement of 148 RB-TnSeq fitness-validated bacterial enzymes (Price-149) across 22 host genomes compared against six enzyme function baselines. Columns report correct Top-1 EC predictions (and per-protein $F_{\max}$ in parentheses) for each standalone predictor, single-predictor METEOR v8 (iteration 2), Stage 2B refinement under minimal glucose and with stoichiometric local-context (SLR + RLGC) probing, full single-predictor GSPA (iteration 3), Stage 0 multi-predictor pooling, and full multi-predictor GSPA (iteration 3). The final column summarizes the Top-1 (and $F_{\max}$) change from standalone baseline to single-predictor and multi-predictor GSPA.

Baseline Predictor	Standalone Base	+ METEOR v8	+ Stage 2B (Min. Glc.)	+ Stage 2B SLR + RLGC	+ GSPA Single-Pred	+ Stage 0 Multi-Pool	+ GSPA Multi-Pred	Single / Multi Gain
CLEAN (2023)	60 (0.5295)	61 (0.5343)	60 (0.5288)	60 (0.5402)	63 (0.5214)	62 (0.6289)	68 (0.6435)	+3 Top-1 / +8 (+0.114)
MAPred (2025)	50 (0.5534)	52 (0.5588)	52 (0.5591)	54 (0.5686)	56 (0.5746)	63 (0.6233)	73 (0.6476)	+6 (+0.021) / +23 (+0.094)
DeepProZyme (2023)	49 (0.5574)	49 (0.5427)	47 (0.5541)	51 (0.5757)	48 (0.5774)	59 (0.6410)	60 (0.6444)	+2 (+0.020) / +11 (+0.087)
EnzBERT (2023)	53 (0.4786)	52 (0.4451)	53 (0.4729)	53 (0.4657)	52 (0.4571)	65 (0.6316)	62 (0.6156)	+0.074 F_1@0.5 / +12 (+0.153)
GraphEC (2024)	18 (0.2634)	18 (0.2250)	21 (0.2834)	21 (0.2851)	20 (0.2783)	59 (0.6174)	67 (0.6368)	+3 (+0.022) / +49 (+0.373)
TopEC (2025)	5 (0.0338)	5 (0.0338)	5 (0.0410)	4 (0.0383)	5 (0.0394)	59 (0.5859)	56 (0.5748)	+0.007 $F_{\max}$ / +54 (+0.552)

metabolic route. We next asked whether attributing these flux-certified reactions competitively to candidate gene loci improves the identification of experimentally validated enzymes.

We evaluated enzyme discovery on 148 divergent bacterial enzymes across 22 host genomes whose physiological roles were established by random-barcoded transposon mutagenesis (Price-149) [59], comparing six standalone enzyme predictors against METEOR v8, single-predictor GSPA, and multi-predictor GSPA (Figure 5b and Table 6). Because 55 of the 148 enzymes catalyze alternative catabolic pathways that are inactive on minimal glucose, single-medium MILP gap-filling lowers $F_{\max}$ on DeepProZyme (0.5574 → 0.5427), EnzBERT (0.4786 → 0.4451), and GraphEC (0.2634 → 0.2250). Operating on a single predictor's proteome output, Stage 2B stoichiometric local-context (SLR + RLGC) probing and single-predictor GSPA improve Top-1 enzyme recovery on MAPred from

50 to 56 proteins ($F_{\max}$ from 0.5534 to 0.5746), on CLEAN from 60 to 63 ($F_{\max}$ to 0.5402 under SLR + RLGC), on DeepProZyme from 49 to 51 ($F_{\max}$ to 0.5774), and on GraphEC from 18 to 21 ($F_{\max}$ to 0.2851). Combining multi-predictor Stage 0 pooling with subsequent locus-level refinement produces both strong pooling gains on lower-recall baselines and additive post-pooling refinement gains on the leading predictors, raising MAPred from 63 after Stage 0 pooling to 73 (49.3% Top-1 accuracy, $F_{\max} = 0.6476$, $p = 6.0 \times 10^{-6}$, exact McNemar test), CLEAN from 62 to 68 ($F_{\max} = 0.6435$), DeepProZyme from 59 to 60 ($F_{\max} = 0.6444$, $p = 0.019$), and GraphEC from 59 to 67 ($F_{\max} = 0.6368$; Table 6). Complementary predictor pooling and competitive stoichiometric locus attribution therefore work additively on the top enzyme predictors to identify divergent, fitness-validated bacterial enzymes. Having evaluated gene-level enzyme recovery, we next examined whether systems-level pathway pooling and coupling predict experimental organismal phenotypes at scale.

2.7 Systems-level annotations predict experimental bacterial phenotypes

We next evaluated whether systems-level pathway pooling and subunit-aware modeling predict experimentally measured bacterial physiology across thousands of strains in BacDive [60, 61]. We tested blind genome-to-phenotype predictions (0% phenotype training labels) across 2,332 carbon-source utilization assays in 529 strains, 17,496 diagnostic enzyme assays in 3,017 type strains, and an extended panel of 50,075 assays for 10 diagnostic enzymes across 6,313 strains (Figure 5d and Table 7). On the 50,075-assay panel, uniform MILP gap-filling (METEOR) lowers overall balanced accuracy from 0.6792 to 0.6631–0.6670 and lowers precision from 0.6442 to 0.6194–0.6285 by adding uncertified diagnostic reactions across strains. Across the 3,017-strain multi-model benchmark, pooling calibrated pathway calls from gapseq, ModelSEED, and CarveMe with unsupervised cross-assay pathway coupling and taxonomic priors in blind Track 1 improves carbon-growth F_1 from 0.7952 (gapseq) to 0.8094 (balanced accuracy = 0.8239, ROC-AUC = 0.8620, $+43/-4$ net improved assays, $p = 3.2 \times 10^{-9}$), diagnostic-enzyme F_1 from 0.7753 to 0.7757 (balanced accuracy = 0.7483, $+74/-52$), and heteromeric urease (`EC 3.5.1.5`) F_1 from 0.5893 to 0.6039 ($+58/-35$, $p = 0.022$; Table 7). Continuous pathway coupling and multi-model calibration therefore improve both binary phenotype accuracy and probability calibration over individual metabolic reconstruction tools. We next examined whether these mechanistic pathway posteriors improve out-of-genus transfer when combined with supervised phenotype classifiers and whether partial phenotypic observations from a strain refine its remaining unassayed pathways.

Under five-fold genus-blocked cross-validation, in which every test strain belongs to a taxonomic genus unseen during training, combining GSPA pathway posteriors with the DiASPora gradient-boosting classifier [62] raises out-of-genus F_1 to 0.8778 (ROC-AUC = 0.9567) on carbon growth, 0.8712 (ROC-AUC = 0.9098) on diagnostic enzymes, and 0.7101 (ROC-AUC = 0.8832) on heteromeric urease, outperforming all individual metabolic reconstructions as well as Traitar [63], BacDive-AI [61], and standalone DiASPora across folds (Table 7 and Supplementary Table S13). In Track 2, where 30% of each strain's experimental assays are revealed as partial phenotypic evidence and the remaining 70% are held out, conditioning GSPA's pathway posteriors via an unsupervised Bayesian update lifts held-out carbon-growth F_1 from 0.7922 (gapseq) to 0.8842 (0.8546 on held-out enzymes), and the hybrid with DiASPora reaches $F_1 = 0.9133$ (ROC-AUC = 0.9770) on carbon growth and $F_1 = 0.8737$ (ROC-AUC = 0.9088) on diagnostic enzymes (Table 7). Mechanistic pathway coupling provides domain-generalizable features for unseen genera and propagates partial experimental observations across coupled metabolic pathways. After deterministic probabilistic and stoichiometric refinement, a small set of ambiguous loci remains, which we examined in Stage 3.

2.8 Guarded language-model reasoning resolves residual ambiguity

Deterministic inference in Stages 1 and 2 resolves the vast majority of proteome annotations, leaving a small fraction of structured anomalies: unresolved pathway gaps, incomplete heteromeric complexes, uncovered essential functions, and borderline ontology conflicts. Across the 88 multi-kingdom proteomes, deterministic anomaly triggers select a mean of 1.84% of proteins per proteome (fewer than 2.5% in every domain of life), bounding the computational cost of invoking a language model. Evaluated across three random seeds on a de-identified benchmark of 120 anomalies from seven prokaryotic and eukaryotic genomes with all gene symbols masked and candidate ordering randomized, a direct zero-shot language model invoked in a single turn without genomic inspection tools or symbolic verification achieves $F_1 = 0.2817$ (precision 0.5882, recall 0.1852, accuracy 0.6083,abstaining on 101 of 120 anomalies). Equipping the model with read-only genomic tools and the symbolic guardrail raises mean F_1 to 0.9115 ± 0.0008 with 0.0% accepted ontology violations, whereas ablating the symbolic guardrail admits a cross-clade violation and reduces F_1 to 0.8515 (Figure 6a,b,d, Table 8, and Supplementary Table S14). Selective triggering therefore confines language-model inference to fewer than 2.5% of proteins, while tool-mediated evidence retrieval and deterministic verification guarantee that accepted proposals satisfy all taxonomic and anatomical axioms. We next tested whether tool-mediated reasoning can resolve multi-candidate metabolic gaps.

When multiple candidate genes in a genomic neighborhood share similar RLGC scores for a missing reaction, resolving the true enzyme requires synthesizing domain

Table 7. Experimental carbon-source growth and diagnostic enzyme phenotype prediction in BacDive compared against metabolic reconstruction and supervised phenotype baselines. Track 1 evaluates blind genome-to-phenotype prediction (0% phenotype labels) against CarveMe, ModelSEED, and gapseq, and 5-fold genus-blocked cross-validation (pooled out-of-fold metrics across unseen genera) against Traitar (L1-SVM), BacDive-AI (Random Forest), and DiASPora (GBDT). Track 2 reveals 30% of each strain's assay rows as partial phenotypic evidence and evaluates predictions on the held-out 70%. Middle rows report the 10-enzyme diagnostic panel (50,075 assays across 6,313 strains) and heteromeric urease (`EC 3.5.1.5`, $n = 2{,}889$ type strains). Runtime is reported per genome after base reconstruction.

Assay Cohort & Protocol	Baseline or Method Pipeline	Strains / Assays	Precision	Recall	F_1 Score	Bal. Acc.	ROC-AUC	McNemar (+/−, p) or Runtime
Track 1: Carbon Growth (Blind & Genus-CV)	CarveMe v1.5.2 (Unsupervised Binary)	529/2,332	0.6935	0.4073	0.5132	0.6233	0.6233	+191/ − 597 (28.4 s)
	ModelSEED v2.6.1 (Unsupervised Binary)	529/2,332	0.8576	0.5200	0.6474	0.7214	0.7214	+186/ − 365 (42.0 s)
	gapseq v1.2 (Unsupervised Binary Baseline)	529/2,332	0.8071	0.7836	0.7952	0.8082	0.8082	Reference (185.0 s)
	GSPA Track 1 (100% Blind Multi-Model + Priors)	529/2,332	0.8390	0.7818	0.8094	0.8239	0.8620	+43/ − 4 ($p = 3.2 \times 10^{-9}$)
	BacDive-AI Random Forest (5-Fold Genus-CV)	529/2,332	0.8433	0.8464	0.8448	0.8530	0.9222	+139/ − 37 ($p < 10^{-14}$)
	DiASPora GBDT (5-Fold Genus-Blocked CV)	529/2,332	0.8475	0.8491	0.8483	0.8564	0.9268	+200/ − 90 ($p < 10^{-10}$)
	Traitar L1-SVM (5-Fold Genus-Blocked CV)	529/2,332	0.8472	0.8673	0.8571	0.8638	0.9185	+204/ − 78 ($p < 10^{-13}$)
	GSPA + DiASPora Hybrid (5-Fold Genus-CV)	529/2,332	0.8839	0.8718	0.8778	0.8848	0.9567	+291/ − 114 ($p < 10^{-18}$)
Track 1: Diagnostic Enzymes (Blind & Genus-CV)	CarveMe v1.5.2 (Unsupervised Binary)	3,017/17,496	0.7739	0.2996	0.4319	0.5774	0.5774	+939/ − 4,977 (28.4 s)
	ModelSEED v2.6.1 (Unsupervised Binary)	3,017/17,496	0.7854	0.4779	0.5943	0.6310	0.6310	+700/ − 3,263 (42.0 s)
	gapseq v1.2 (Unsupervised Binary Baseline)	3,017/17,496	0.8393	0.7204	0.7753	0.7461	0.7461	Reference (185.0 s)
	GSPA Track 1 (100% Blind Multi-Model + Priors)	3,017/17,496	0.8425	0.7188	0.7757	0.7483	0.7465	+74/ − 52 ($p = 0.0609$)
	BacDive-AI Random Forest (5-Fold Genus-CV)	3,017/17,496	0.8391	0.8527	0.8458	0.7911	0.8593	+1,481/ − 317 ($p < 10^{-100}$)
	DiASPora GBDT (5-Fold Genus-Blocked CV)	3,017/17,496	0.8602	0.8499	0.8550	0.8108	0.8699	+1,920/ − 509 ($p < 10^{-100}$)
	Traitar L1-SVM (5-Fold Genus-Blocked CV)	3,017/17,496	0.8402	0.8794	0.8593	0.8014	0.8715	+2,204/ − 790 ($p < 10^{-100}$)
	GSPA + DiASPora Hybrid (5-Fold Genus-CV)	3,017/17,496	0.8652	0.8774	0.8712	0.8256	0.9098	+2,568/ − 843 ($p < 10^{-100}$)
10-EC Diagnostic Panel (50,075 Assays)	DeepProZyme (2023) Standalone Baseline	6,313/50,075	0.6442	0.8105	0.7178	0.6792	0.6792	Reference baseline
	+ METEOR a025 / v7 (Minimal-Glucose MILP)	6,313/50,075	0.6285	0.8285	0.7148	0.6670	0.6670	Over-gap-filling drop
Heteromeric Urease (`EC 3.5.1.5`)	gapseq v1.2 (Unsupervised Binary)	2,889 strains	0.6184	0.5629	0.5893	0.7461	0.7461	Reference (185.0 s)
	GSPA Track 1 (100% Blind Multi-Model + Priors)	2,889 strains	0.6596	0.5569	0.6039	0.7495	0.7510	+58/ − 35 ($p = 0.022$)
	GSPA + DiASPora Hybrid (5-Fold Genus-CV)	2,889 strains	0.7330	0.6886	0.7101	0.8181	0.8832	+501/ − 219 ($p < 10^{-26}$)
Track 2: Carbon Held-Out 70% (30% Revealed)	gapseq v1.2 (Unsupervised Binary)	529/1,641	0.8059	0.7789	0.7922	0.8049	0.8049	Reference (185.0 s)
	DiASPora GBDT (Trained on 30% Revealed)	529/1,641	0.8727	0.8985	0.8854	0.8901	0.9575	+170/ − 33 ($p < 10^{-22}$)
	GSPA Track 2 (Unsupervised Bayesian Anchor)	529/1,641	0.8290	0.9473	0.8842	0.8856	0.9536	+203/ − 78 ($p < 10^{-13}$)
	GSPA Track 2 + DiASPora Hybrid	529/1,641	0.9127	0.9139	0.9133	0.9175	0.9770	+241/ − 58 ($p < 10^{-26}$)
Track 2: Enzymes Held-Out 70% (30% Revealed)	gapseq v1.2 (Unsupervised Binary)	3,017/12,316	0.8376	0.7188	0.7737	0.7446	0.7446	Reference (185.0 s)
	GSPA Track 2 (Unsupervised Bayesian Anchor)	3,017/12,316	0.8074	0.9078	0.8546	0.7755	0.8736	+1,707/ − 851 ($p < 10^{-62}$)
	GSPA Track 2 + DiASPora Hybrid	3,017/12,316	0.8622	0.8856	0.8737	0.8262	0.9088	+1,825/ − 564 ($p < 10^{-100}$)

Table 8. Stage 3 neuro-symbolic disambiguation compared against deterministic, zero-shot LLM, threshold, and supervised baselines across 120 de-identified anomalies and 50 ambiguous metabolic gaps. All gene symbols are masked and candidate order is hash-shuffled. The 120 anomalies comprise 54 true-positive promotable gaps, 36 score-matched decoys, and 30 ontology conflicts (15 genuine cross-clade conflicts and 15 adversarial traps). Supervised classifiers are evaluated via 5-fold cross-validation using either 3 pre-tool scalars or 8 post-tool features combined with the symbolic guardrail. Latency reports concurrency-normalized wall-clock seconds per anomaly at batch concurrency 8 (raw HTTP latency in parentheses).

Resolution Method	Precision	Recall	F_1 Score	Accuracy	Prom. / Relax.	Violations Accepted	LRO-50 Hit@1 (MRR)	Latency / Tools
Stage 1 + Stage 2 Deterministic Only	0.0000	0.0000	0.0000	0.5500	0/0	0.0% (0/120)	34.0% (0.6020)	0.00 s / 0.00
Direct Zero-Shot LLM (No Tools, No Guardrail)	0.5882	0.1852	0.2817	0.6083	17/1	0.0% (0/120)	42.0% (0.6133)	0.91 s (7.24 s) / 0.00
Posterior Threshold (≥ 0.35) + Guardrail	0.5686	0.5370	0.5524	0.7083	29/30	0.0% (0/120)	44.0% (0.6233)	< 0.01 s / 0.00
Logistic Regression (5-Fold CV, 3 Pre-Tool Features)	0.8571	0.6667	0.7500	0.8000	42/0	2.5% (3/120)	94.0% (0.9667)	< 0.01 s / 0.00
GBDT (5-Fold CV, 3 Pre-Tool Features)	0.9111	0.7593	0.8283	0.8583	45/0	0.0% (0/120)	90.0% (0.9383)	< 0.01 s / 0.00
Stage 3 Fast 1-Step Mode (Max 2 Turns)	1.0000	0.1667	0.2857	0.6250	9/4	0.0% (0/120)	36.0% (0.5800)	2.47 s (19.8 s) / 5.01
Stage 3 Multi-Turn Without Symbolic Guardrail	0.9149	0.7963	0.8515	0.8833	47/0	0.8% (1/120)	82.0% (0.8740)	3.72 s (29.7 s) / 5.07
Stage 3 Multi-Turn + Guardrail (3-Seed Mean ± s.d.)	0.9722 ± 0.009	0.8581 ± 0.009	0.9115 ± 0.001	0.9278 ± 0.004	46.3/13.7	0.0% (0/120)	$82.7 \pm 0.9\%$ (0.879)	4.07 s (32.5 s) / 5.74
Logistic Regression (5-Fold CV, 8 Tool Features + Guardrail)	0.9792	0.8704	0.9216	0.9333	48/1	0.0% (0/120)	96.0% (0.9767)	< 0.01 s / 0.00
Random Forest (5-Fold CV, 8 Tool Features + Guardrail)	0.9286	0.9630	0.9455	0.9500	56/2	0.0% (0/120)	92.0% (0.9517)	< 0.01 s / 0.00
GBDT (5-Fold CV, 8 Tool Features + Guardrail)	0.9800	0.9074	0.9423	0.9500	50/0	0.0% (0/120)	92.0% (0.9500)	< 0.01 s / 0.00
MLP (5-Fold CV, 8 Tool Features + Guardrail)	0.9811	0.9630	0.9720	0.9750	53/0	0.0% (0/120)	92.0% (0.9507)	< 0.01 s / 0.00

architecture, operon structure, and pathway stoichiometry. Across 50 unique-protein ambiguous leave-reaction-out gaps in *E. coli* K-12 where Stage 1 RLGC alone ranks the true enzyme first in 34.0% of cases (42.0% for a direct zero-shot LLM and 44.0% for posterior thresholding with the guardrail), Stage 3 multi-turn re-ranking lifts Top-1 recovery to a three-seed mean of $82.7 \pm 0.9\%$ ($p = 2.8 \times 10^{-7}$, exact McNemar test; Figure 6c and Table 8). Tool-mediated evidence synthesis resolves ties among candidate enzymes after deterministic priors have narrowed the search space. Because a supervised tabular classifier can also be trained on numerical features extracted by the same tools, we compared zero-shot guarded reasoning against cross-validated classifiers to establish the operating regime of each method.

We compared the zero-shot guarded agent against logistic regression, random forest, gradient-boosted decision tree (GBDT), and multilayer perceptron classifiers evaluated by five-fold cross-validation on either three pre-tool summary scalars or eight post-tool genomic and domain features paired with the symbolic guardrail (Table 8). On the 120 anomalies, the zero-shot agent ($F_1 = 0.9115$) outperforms all classifiers restricted to pre-tool scalars ($F_1 = 0.7500$–0.8283), while supervised classifiers trained on the eight post-tool features reach up to $F_1 = 0.9720$ on binary anomaly promotion and 96.0% Hit@1 on the 50 metabolic gaps at sub-millisecond inference latency. On borderline ontology conflicts, the zero-shot agent proposes a mean of 13.7 valid ancestral GO relaxations compared with 1–2 for the post-tool classifiers, which select among fixed candidate labels. Supervised post-tool classifiers are therefore preferable for high-throughput binary promotion and candidate ranking when labeled anomalies are available, whereas zero-shot guarded reasoning provides zero-label operation and multi-class ancestral GO relaxation (see Supplementary Section S6 and Supplementary Table S16 for an application to five desert extremophile isolates from the Empty Quarter). Having validated all four pipeline stages across controlled holdouts and experimental phenotype benchmarks, we next scaled whole-genome inference across 72,533 genomes from the AllTheBacteria

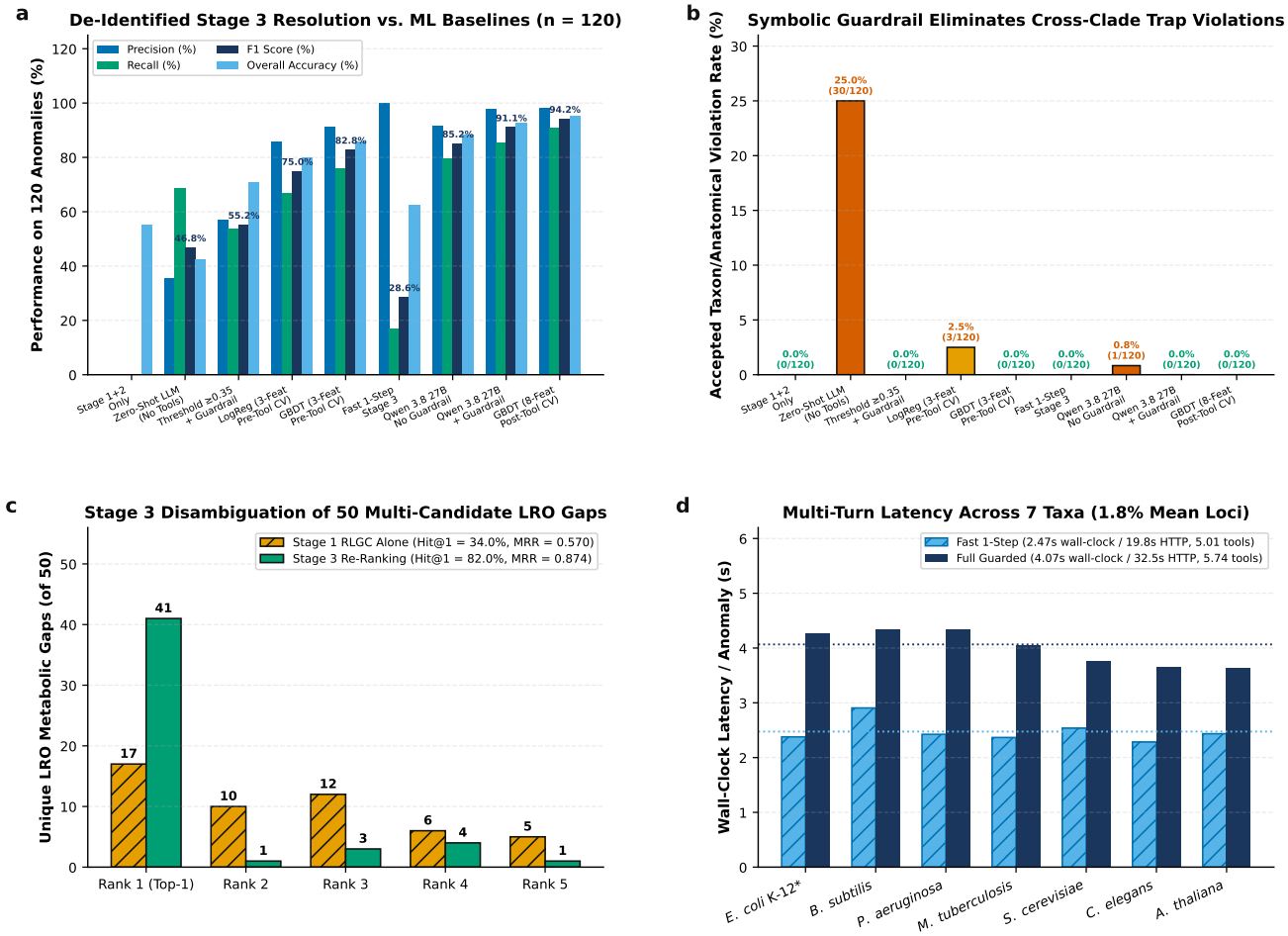


Figure 6. Evaluation of Stage 3 selective neuro-symbolic reasoning on de-identified benchmarks. (**a**) Precision, recall, F_1, and resolution accuracy across 120 de-identified anomalies (30 pathway gaps, 30 incomplete complexes, 30 borderline essential functions, and 30 borderline ontology conflicts) comparing deterministic Stages 1 and 2, direct zero-shot LLM without tools or guardrail, posterior thresholding (≥ 0.35) with the symbolic guardrail, 5-fold cross-validated supervised classifiers on pre-tool and post-tool features, fast 1-step Stage 3, unguarded multi-turn Stage 3, and full guarded multi-turn Stage 3 (3-seed mean). (**b**) Rate of accepted taxonomic and anatomical violations across configurations. (**c**) Rank distribution of the true enzyme across 50 unique-protein ambiguous LRO metabolic gaps before and after Stage 3 re-ranking. (**d**) Concurrency-normalized wall-clock latency per anomaly across seven prokaryotic and eukaryotic species (mean 1.84% of proteome loci triggered).

resource [54] to determine what comparative laws govern sequence dark matter, complex stoichiometry, and pathway evolution across the bacterial domain.

2.9 Pan-phylum laws of bacterial dark matter, complex stoichiometry, and non-orthologous displacement across 72,533 genomes

To evaluate whole-genome inference across bacterial and archaeal diversity, we implemented a two-stage sequence deduplication pipeline that hashes identical coding sequences via SHA-256 (4.27-fold compression across 315,671,487 proteins in 78,035 assemblies to 73,866,871 unique sequences) and clusters them in linear time via `MMseqs2 linclust` [64] at 90% sequence identity and 80% coverage (8.58-fold total compression to 36,804,001 cluster representatives) before scoring representatives with GSPA-Seq and executing whole-genome integration (`-batch-manifest`) across 72,533 complete genomes spanning 292,133,146 coding sequences, 11,632 species, and 37 phyla (Figure 7a, Supplementary Section S7, and Supplementary Table S17). Across all 292,133,146 coding sequences, 119,647,577 (40.96%) lie in the sequence dark-matter regime (pident $<$ 30% to Swiss-Prot), of which 102,673,326 (85.81%) have zero detectable BLAST alignment (pident $=$ 0.0% at $E \leq 10^{-3}$) and 16,974,251 (14.19%) have remote alignments (0% $<$ pident $<$ 30%), while an additional 78,843,303 sequences (26.99%) lie in the twilight zone (30% $\leq$ pident $<$ 50%). Sequence dark matter varies nearly twofold across phyla, rising from 36.51% in Actinomycetota (135.8 million coding sequences) and 40.42% in Pseudomonadota (86.9 million) to 60.82%–66.64% in environmental, predatory, and symbiotic phyla including Planctomycetota (66.64%), Bdellovibrionota_B (66.02%), Bacteroidota (64.49%), Armatimonadota (62.58%), Fibrobacterota (62.52%), Methanobacte-

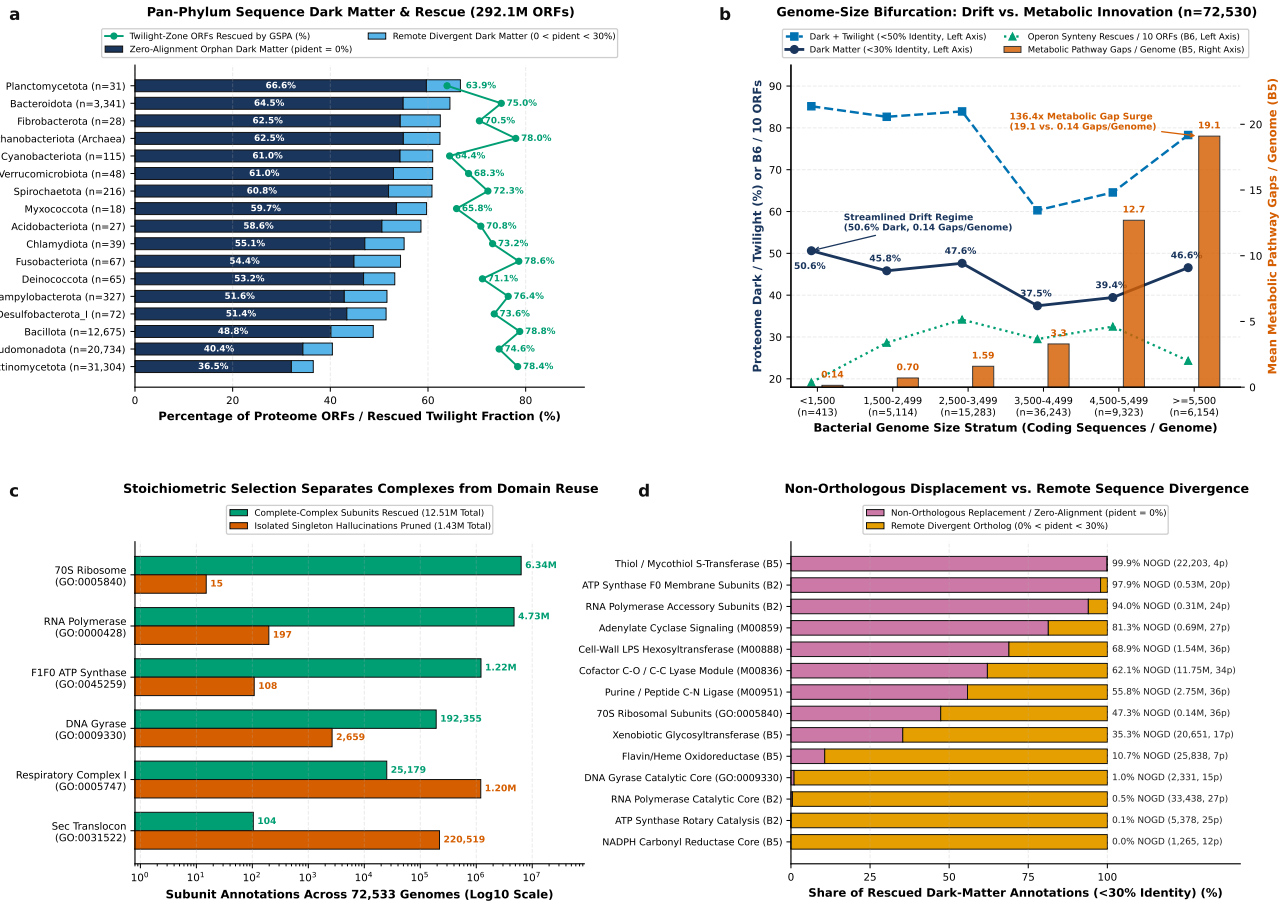


Figure 7. Pan-phylum laws of bacterial dark matter, genome-size bifurcation, stoichiometric selection, and non-orthologous gene displacement across 72,533 **complete genomes (**292.1 **million coding sequences,** 11,632 **species,** 37 **phyla).** (**a**) Proportion of proteome coding sequences in the sequence dark-matter regime ($< 30\%$ pairwise identity to Swiss-Prot, partitioned into zero-alignment orphans with pident $= 0\%$ and remote divergent homologs with $0\% <$ pident $< 30\%$) across major bacterial and archaeal phyla, paired with the fraction of twilight-zone proteins ($30\% \leq$ pident $< 50\%$) rescued by whole-genome GSPA inference. (**b**) Genome-size bifurcation across 72,530 bacterial genomes stratified into six ORF-count regimes: sequence dark matter follows a U-shaped trajectory driven by sequence drift in streamlined genomes ($<$ 1,500 ORFs; 50.62% dark matter, 0.14 metabolic gaps per genome), whereas metabolic pathway gaps (B_5) surge 136.4-fold per genome in large environmental genomes ($\geq$ 5,500 ORFs; 19.10 gaps per genome; Spearman $\rho = 0.5627, p < 10^{-300}$). (**c**) Stoichiometric selection across six obligate prokaryotic complexes ($\log_{10}$ scale), comparing 12.51 million subunit annotations rescued by whole-genome complex coherence (B_2) in genomes encoding complete partner sets against 1.43 million isolated singleton subunit predictions pruned by Stage 2A (predominantly soluble NADH-module domain reuse cross-hitting Respiratory Complex I `GO:0005747` and Sec translocon `GO:0031522`). (**d**) Partitioning of rescued dark-matter annotations ($<$ 30% identity) into zero-alignment non-orthologous replacements (pident $= 0\%$) and remote divergent orthologs ($0\% <$ pident $< 30\%$) across macromolecular complex subunits and metabolic modules/pathways (p denotes the number of phyla exhibiting dark-matter rescue).

riota (62.45%), Gemmatimonadota (61.50%), Cyanobacteriota (60.96%), Verrucomicrobiota (60.96%), Bdellovibrionota (60.92%), and Spirochaetota (60.82%). Across all 37 phyla, whole-genome priors rescue 51,375,006 dark-matter proteins (42.94%, comprising 39,253,161 zero-alignment orphans [76.41%] and 12,121,845 remote homologs [23.59%]) and 60,674,994 twilight-zone proteins (76.96%) above the 0.50 posterior threshold (353,243,657 promoted dark-matter protein–term pairs: 88.58% from co-directional operon synteny B_6, 11.40% from pathway and complex coherence B_2, and 0.020% from surgical metabolic gap-filling B_5), while pre-propagation taxonomic (B_3) and anatomical (B_4) enforcement eliminates 47,739,228 direct taxonomic and 2,388,329,971 propagated anatomical cross-kingdom violations (2,436,069,199 total violations, or 33,586 per genome). Conserved operon organization and macromolecular stoichiometry therefore persist across phyla even when primary amino-acid sequences diverge beyond alignment thresholds. We next asked how sequence dark matter and metabolic pathway gaps scale with bacterial genome size.

Stratifying the 72,530 bacterial genomes across six genome-size regimes ($<$ 1,500 to $\geq$ 5,500 ORFs; Figure 7b and Supplementary Table S18) reveals a functional bifurcation between sequence dark matter and metabolic network gaps. Sequence dark matter follows a U-shaped curve across genome size: streamlined and host-associated bacteria ($<$ 1,500 ORFs; 413 genomes across 238 species and 17 phyla, mean 1,152.3 ORFs) carry the highest dark-matter fraction (50.62% dark matter and 34.55% twilight zone, totaling 85.17% of the proteome below 50% iden-

tity), which drops to a minimum of 37.46% dark matter (22.83% twilight zone) in modal bacterial generalists (3,500–4,499 ORFs; 36,243 genomes across 3,123 species) and rises back to 46.57% dark matter (31.72% twilight zone) in large environmental and secondary-metabolite genomes ($\geq$ 5,500 ORFs; 6,154 genomes across 2,549 species, mean 6,609.9 ORFs). In contrast, metabolic pathway gaps (B_5) increase monotonically with genome size (Spearman $\rho = 0.5627, p < 10^{-300}$), surging 136.4-fold per genome from 0.14 gaps per genome (0.108 per 1,000 ORFs) in $<$ 1,500-ORF genomes to 12.70 in 4,500–5,499-ORF genomes and 19.10 (2.876 per 1,000 ORFs, a 26.6-fold density increase) in $\geq$ 5,500-ORF genomes. Streamlined bacteria shed peripheral catabolic pathways while retaining core informational and bioenergetic complexes whose subunits undergo rapid sequence drift inside conserved operons (1.914 B_6 rescues per ORF), whereas large environmental genomes accumulate lineage-specific catabolic and secondary biosynthetic operons that introduce stoichiometrically unannotated reaction steps. Because macromolecular complexes are retained across these regimes while metabolic enzymes expand, we next examined how whole-genome stoichiometric selection distinguishes intact multi-subunit complexes from modular domain reuse.

Comparing complex coherence promotions (B_2) against Stage 2A singleton pruning across 72,533 genomes separates conserved multi-subunit complexes from standalone enzymes that reuse homologous catalytic domains (Figure 7c and Supplementary Table S18). In genomes encoding the complete partner subunit complement, whole-genome coherence rescues 12,506,795 divergent subunit annotations across six obligate prokaryotic complexes, led by the 70S ribosome (`GO:0005840`, 6,339,928 rescued subunits versus 15 pruned singletons), DNA-directed RNA polymerase (`GO:0000428`, 4,733,571 versus 197), proton-transporting F_1F_0 ATP synthase (`GO:0045259`, 1,215,204 versus 108), and DNA gyrase (`GO:0009330`, 192,355 versus 2,659). Conversely, Stage 2A prunes 1,426,276 isolated singleton predictions where per-protein predictors assign an obligate complex term in the absence of its partner subunits, concentrated in respiratory NADH dehydrogenase Complex I (`GO:0005747`, 1,202,957 pruned singletons across Actinomycetota [615,599], Pseudomonadota [397,602], Bacillota [68,387], and Bacteroidota [43,304], alongside 25,179 rescued subunits in genomes encoding the complete `nuoA-N` complex) and the Sec protein export translocon (`GO:0031522`, 220,519 pruned singletons). Per-protein homology and language-model predictors assign respiratory Complex I terms to standalone fermentative [4Fe-4S] flavoproteins, bidirectional hydrogenases, and soluble Rossmann-fold oxidoreductases that share the soluble N-module NADH-binding domains (`NuoE/F/G`) while lacking the membrane proton-pumping core (`NuoA/H/J/K/L/M/N`); whole-genome stoichiometric enforcement eliminates these false-positive complex assignments while rescuing divergent subunits of intact bioenergetic complexes. Having separated intact complexes from domain reuse, we asked whether the dark-matter proteins rescued inside complexes, operons, and pathways represent remote sequence orthologs or non-orthologous gene displacements.

Partitioning rescued dark-matter annotations ($<$ 30% identity) into zero-alignment replacements (pident = 0.0%) and remote divergent orthologs (0% $<$ pident $<$ 30%) uncovers a division of evolutionary modes between catalytic cores and peripheral or adaptive functions (Figure 7d and Supplementary Table S18). Within multi-subunit complexes and core redox pathways, catalytic active-site subunits evolve primarily through gradual sequence divergence while retaining alignment traces: 99.95% of dark-matter rescues for ATP synthase rotational proton transport (`GO:0046933`, 5,375 of 5,378 across 25 phyla), 99.53% for RNA polymerase core transcription (`GO:0006351`, 33,280 of 33,438 across 27 phyla), 99.00% for DNA gyrase topological catalysis (`GO:0006265`, 2,308 of 2,331 across 15 phyla), 100.0% for Complex I electron-transfer catalysis (`GO:0003954` and `GO:0042773`), and 100.0% for NADPH carbonyl reductases (`GO:0004090`, 1,265 across 12 phyla) and secondary-metabolite monooxygenases (`GO:0097621`, 795 across 13 phyla) carry remote sequence alignments (0% $<$ pident $<$ 30%). In contrast, short membrane-embedded and peripheral complex subunits (including 97.86% of dark-matter F_0 membrane subunit rescues in ATP synthase `GO:0045259`, 520,762 of 532,135 across 20 phyla, and 93.99% of small accessory ω/δ subunit rescues in RNA polymerase `GO:0000428`, 292,899 of 311,636 across 24 phyla, as well as 82.74% of all co-directional operon synteny promotions B_6, 258,892,062 of 312,893,381) diverge completely beyond sequence alignment (pident = 0.0%) yet remain conserved within their canonical operon slots. Similarly, across metabolic modules (B_2, 49.08% overall zero-alignment) and pathway gap-fills (B_5, 45.19% overall zero-alignment), non-orthologous gene displacement (pident = 0.0%) predominates in lineage-specific thiol and mycothiol S-transferases (`GO:0004364`, 99.91% across 4 phyla), adenylate cyclase signaling (`KEGG:M00859`, 81.34% across 27 phyla), cell-wall lipopolysaccharide hexosyltransferases (`KEGG:M00888`, 68.89% across 36 phyla), cofactor C–O/C–C lyases (`KEGG:M00836`, 62.09% across 34 phyla), purine and peptide C–N ligases (`KEGG:M00951`, 55.80% across 36 phyla), and xenobiotic glycosyltransferases (`GO:0015020`, 35.34% across 17 phyla). Whole-genome inference therefore unifies the recovery of remote catalytic orthologs with the systematic mapping of non-orthologous gene displacements across bacterial phyla.

3 Discussion

Protein function prediction has traditionally been formulated as a per-protein classification task in which each sequence or structure is scored independently [1, 20, 21, 24, 25, 27, 29]. GSPA reformulates proteome annotation as staged inference over the complete genome of an organism, and our evaluation shows that three complementary

drivers account for its performance. First, calibrated pooling of sequence homology, domain profiles, protein language models, and structural similarity establishes a strong per-protein probability distribution in Stage 0, instantiated as our standalone per-protein stacking integrator GSPA-Seq. Second, whole-genome priors and deterministic cross-ontology enforcement in Stages 1 and 2A update these probabilities using organism-wide invariants over operons, core essential functions, obligate heteromeric complexes, and taxonomic and anatomical consistency. Third, stoichiometric flux modeling, competitive gene-locus attribution, and selective ontology-guarded reasoning in Stages 1, 2B, and 3 connect metabolic network requirements and multi-source genomic evidence to individual gene loci. Across all six splits of our whole-genome withheld CAFA evaluation (Table 1), integrating complete genome assemblies improves both information-accretion weighted f_w and unweighted $F_{\max}$ over official pre-trained and retrained sequence, domain, convolutional, graph neural network, transformer, semantic entailment, and protein language model baselines (including DeepFRI [27], DeepGraphGO [20], TALE+ [21], SPROF-GO [24], DeepGO-SE [25], and DPFunc [29]) as well as over GSPA-Seq, while eliminating all taxonomic and anatomical violations.

Stage 0 and GSPA-Seq build directly on the multi-source ensemble paradigm introduced by GOLabeler [30] and NetGO [31, 32], which showed that integrating sequence alignment, domain signatures, motifs, network associations, and language-model embeddings outperforms individual representations in CAFA evaluations. GSPA extends this principle to genome-scale inference by calibrating raw predictor outputs via Platt scaling, grouping correlated evidence channels so that shared homology or embedding signals are counted once, and pooling across groups via Noisy-OR into an anchored log-odds prior $L^{(0)}$. Because every subsequent Stage 1 update is added to $L^{(0)}$, sequence homology, domain profiles, protein language models, and structural similarity remain the foundation of the framework. Conserved proteins retain their high-confidence Stage 0 assignments, and the recall of the Stage 0 candidate pool defines the space of candidate functions that genome-scale priors refine.

Exploiting whole-genome information in Stages 1 and 2A is timely because long-read sequencing and modern repeat-graph assemblers now routinely recover complete or near-complete chromosomes for both cultivated isolates and metagenome-assembled genomes [10–12]. Complete assemblies expose two distinct scales of genomic organization, and our whole-genome CAFA ablations (Table 1) and pan-phylum AllTheBacteria analysis across 72,533 genomes [54] (Figure 7 and Supplementary Tables S17 and S18) reveal both how each mechanism contributes to functional accuracy and what biological principles govern bacterial genomes at scale. Comparing the two whole-genome temporal splits shows that pre-t_0 experimental annotations on neighboring genes along the same chromosome provide contextual anchors for newly characterized genes, lifting whole-genome f_w from 0.4455 in de novo `WG_Time_Withheld` to 0.4996 in `WG_Time_Rollover` (+0.0541 f_w). Across 72,533 genomes from 37 phyla, whole-genome inference uncovers four comparative laws of bacterial functional architecture: (i) conserved co-directional operon grammar and complex stoichiometry rescue 42.9% (51.4 million) of sequence dark-matter proteins and 77.0% (60.7 million) of twilight-zone proteins, resolving the apparent core auxotrophy of high-dark-matter phyla such as Planctomycetota (66.6% dark matter), Bacteroidota (64.5%), Cyanobacteriota (61.0%), and Spirochaetota (60.8%); (ii) bacterial dark matter bifurcates across genome size into a streamlined drift regime ($<$ 1,500 ORFs; 50.6% dark matter, 0.14 metabolic gaps per genome) and a large-genome metabolic innovation regime ($\geq$ 5,500 ORFs; 19.10 gaps per genome, a 136.4-fold surge); (iii) stoichiometric selection separates 12.51 million subunits of intact obligate complexes from 1.43 million isolated singletons caused by modular reuse of soluble Complex I NADH-binding domains in fermentative oxidoreductases; and (iv) catalytic complex and redox cores evolve via remote sequence divergence (99.0%–100.0% alignment retention), whereas peripheral membrane subunits (94.0%–97.9% zero-alignment) and lineage-specific thiol, cell-wall, and cofactor enzymes (55.8%–99.9% zero-alignment) evolve via non-orthologous gene displacement. Enforcing taxonomic and anatomical compatibility on leaf claims prior to upward GO ancestor closure simultaneously reduces zero-violation semantic distance $S_{\min}$ and improves f_w and $F_{\max}$ across all six whole-genome CAFA splits (Table 1), eliminates 2.44 billion cross-kingdom violations across AllTheBacteria, and preserves conserved tissue-level biology in multicellular eukaryotes (Table 4), while operon synteny lifts top-ranked subunit identification in leave-subunit-out experiments from 56.0% to 93.7% (Table 3).

Our third driver advances functional inference beyond sequence homology by integrating systems biology with genomic architecture, addressing the specific mathematical limitations we identified in our earlier method METEOR [47]. METEOR coupled metabolic gap-filling with enzyme prediction through a reaction-indicator MILP that assigned negative objective costs to candidate reactions above an addition threshold and boosted all candidate genes of a selected reaction uniformly. Our 108-genome stoichiometric audit shows that negative addition costs incentivize the solver to accumulate 83.7% zero-flux reaction padding (Figure 5a), while uniform gene boosting elevates non-cognate paralogs and isolated complex subunits. GSPA resolves both issues by coupling reaction-local genomic context (RLGC) in Stage 1 with role-aware disjunctive normal form (DNF) aggregation of isozymes and required subunits, positive-cost lexicographic pFBA [53], and competitive locus responsibility against an explicit unassigned-gene hypothesis in Stage 2B. Under positive addition costs, every added reaction carries steady-state flux in an elementally balanced network, and competitive responsibility concentrates the backward log-odds update on

the candidate gene supported by operon neighborhood and pathway synteny, improving both whole-genome CAFA accuracy (B_5 + Stage 2B in Table 1) and gene-level discovery on RB-TnSeq fitness-validated bacterial enzymes [59] (Table 6). At the organism level, pooling calibrated metabolic reconstructions with unsupervised cross-assay pathway coupling and subunit-complete enzyme rules accurately predicts carbon-source growth and diagnostic enzyme phenotypes across unseen genera in BacDive (Table 7).

Finally, Stage 3 shows how tool-using language models and formal symbolic reasoning can be combined selectively at proteome scale. By restricting invocation to deterministic post-Stage-2 anomaly triggers that select fewer than 2.5% of proteins per proteome, GSPA restricts language-model inference to ambiguous loci while leaving conserved genes resolved by Stages 0–2 untouched. Within each triggered anomaly, the language model synthesizes heterogeneous operon, domain, pathway, and ontology evidence through read-only tools, while a deterministic symbolic guardrail verifies every proposed resolution against GO, NCBI Taxonomy, and the GO–Uberon/PO/CL bridge, mirroring neuro-symbolic verification principles in biomedical curation [51, 52]. Our comparison with five-fold cross-validated tabular classifiers delineates clear deployment regimes: supervised classifiers trained on post-tool features and paired with the symbolic guardrail provide the highest binary promotion accuracy at sub-millisecond latency when labeled anomalies exist, whereas the zero-shot guarded agent operates without anomaly training labels and performs multi-class traversal to valid ancestral GO terms.

3.1 Limitations

Our evaluation design establishes how genome-scale priors and stoichiometric refinement build upon a per-protein candidate set, and four aspects of this design define concrete next experiments. First, on well-annotated reference genomes, roughly 85% of proteins already carry high-confidence homologs in Stage 0 ($P_0 \geq 0.90$), so Stage 0 governs baseline protein-centric performance while Stages 1–3 concentrate their updates on twilight-zone proteins, newly characterized rollover genes, pathway gaps, incomplete complexes, and taxonomic conflicts. Because the composite GAEF score evaluates the same structural invariants over essentiality, pathway and complex coherence, and taxonomic consistency that Stages 1 and 2 optimize, we report GAEF as an internal structural audit score paired with independent held-out whole-genome CAFA, EC, leave-reaction-out, leave-subunit-out, Price-149, and BacDive evaluations, and we evaluate both endogenous unperturbed contradictions (1,009 eukaryotic violations) and synthetic cross-clade stress-test injections. Similarly, in leave-reaction-out evaluation, unconditional enzyme recovery is bounded by whether Stage 0 places the true gene into the candidate pool (70.7% pool coverage in *E. coli* and 52.8% in *B. subtilis*), and Track B uses released ESM-2 weights pretrained on UniRef50, making strict Track A cosine masking the primary test of de novo context recovery. In addition, whereas our six-split whole-genome withheld CAFA evaluation benchmarks complete chromosomal assemblies across six reference species (20,697 ORFs), external community challenge target lists (CAFA6 and Holdout-125) distribute isolated target sequences without source genome coordinates, evaluating Stages 0, 2A, and 3, while the leave-subunit-out benchmark samples domain-matched decoys from non-operonic loci. Expanding Stage 0 candidate generation with remote structural fold search, evaluating protein language models trained under strict chronological splits, sampling leave-subunit-out decoys from within-operon paralogs, and linking community challenge target lists to their complete source genome assemblies will directly measure how increased candidate recall and full chromosomal context affect these regimes.

Second, Stage 2B models steady-state, mass-balanced metabolic flux over reactions mapped in the sanitized universal network under specified growth media, and Stage 2A enforces anatomical constraints over the 2,715 GO terms that currently carry explicit Uberon, PO, or CL logical definitions. Experimental growth and enzyme phenotypes in Price-149 and BacDive also depend on transcriptional regulation, transporter substrate specificity, and cofactor synthesis, while anatomical relaxation in unicellular fungi and non-vascular plants reflects the granularity of tissue-level ancestor annotations in reference databases. Integrating condition-specific transcriptomic expression priors and automated transporter substrate predictors into Stage 2B, alongside expanding cross-ontology logical definitions across additional GO biological processes and eukaryotic clades, will extend stoichiometric and ontological coverage to broader physiological states.

Third, Stage 3 was evaluated using a single locally hosted open-weight instruction model across three random seeds, and the 568 high-confidence pathway gap fills across 97 uncultured MAGs (as well as the extremophile pathway completions in the Empty Quarter cohort) represent ranked computational hypotheses. Benchmarking additional open-weight model families on the de-identified 120-anomaly suite, evaluating zero-shot Stage 3 against supervised post-tool classifiers under a leave-clade-out anomaly split, implementing an adaptive router that dispatches binary in-distribution promotions to the >100-fold faster supervised classifier while reserving multi-turn language-model reasoning for novel clades and ancestral GO relaxations, and validating top-ranked MAG and extremophile enzyme candidates via heterologous expression and mutant fitness assays will establish both computational generality and experimental confirmation.

4 Methods

4.1 Problem formulation and four-stage architecture

Let $\mathcal{P} = \{p_1, \ldots, p_N\}$ denote the protein-coding genes of a target genome with NCBI Taxonomy identifier τ_{tax}, and let $\mathcal{K} = \{(c_i, s_i, e_i, \zeta_i)\}_{i=1}^{N}$ denote their genomic coordinates (contig c_i, start s_i, end e_i, and strand $\zeta_i \in \{+, -\}$). Let $\mathcal{T}$ denote the Gene Ontology (GO) directed acyclic graph under the true-path relation $t \sqsubseteq a$ (`is_a` and `part_of`), with ancestor closure $\text{Anc}(t)$, $\mathcal{M}$ the universal metabolic reaction network with stoichiometric matrix $\mathbf{S}$ and reaction–EC–GO mappings, and $\mathcal{O}$ the formal taxonomic and cross-ontology anatomical axiom set. GSPA estimates a proteome-wide posterior probability matrix $\mathbf{P} \in [0,1]^{N \times |\mathcal{T}|}$ (and corresponding EC probabilities in enzyme benchmarks) conditioned jointly on $\mathcal{K}$, τ_{tax}, $\mathcal{M}$, and $\mathcal{O}$ across four sequential stages (Figure 1): Stage 0 calibrates and pools per-protein evidence channels; Stage 1 performs iterative whole-proteome Bayesian log-odds refinement; Stage 2A applies deterministic symbolic constraint enforcement with semantic ancestor relaxation while Stage 2B performs positive-cost lexicographic parsimonious flux balance analysis (pFBA) and competitive gene-locus attribution; and Stage 3 applies selective tool-mediated neuro-symbolic reasoning verified by a symbolic ontology guardrail. Supplementary Table S1 summarizes all eight benchmark suites, cohorts, Stage 0 channel configurations, and active pipeline stages.

4.2 Stage 0: multi-channel evidence calibration and correlation-grouped Noisy-OR pooling

Stage 0 ingests functional claims from either a standalone predictor (such as CLEAN [26], MAPred [65], DeepProZyme [57], EnzBERT [56], GraphEC [66], or TopEC [67]) or up to seven complementary per-protein channels $e \in \mathcal{E}_{\text{ch}}$: (1) DIAMOND v2.1.8 sequence alignment against experimentally annotated Swiss-Prot [13]; (2) HMMER 3.4 profile HMM search against Pfam-A mapped via Pfam2GO [16]; (3) InterProScan v5.68-100.0 signature integration [17]; (4) ESM-2 (650M or 3B) embedding k-nearest-neighbor and multilayer perceptron transfer [22]; (5) ProtT5-XL-U50 embedding k-nearest-neighbor transfer [23]; (6) neural sequence classifiers (ProteInfer [68], 16-kernel DeepGOCNN and DeepGOPlus [19], DeepGOWeb [69], and DeepGO-SE [25]); and (7) structural similarity transfer via FoldSeek [28] and ProstT5 3Di structural tokens [70].

Each raw claim $c = (p, f, s(c), e(c))$ with raw score $s(c) \in [0,1]$ is mapped to a calibrated probability via Platt logistic scaling, $\tilde{q}(c) = \text{clip}\big(\sigma_{\log}(a_{e(c)}\, s(c) + b_{e(c)}),\ \varepsilon_{\text{cal}},\ 1 - \varepsilon_{\text{cal}}\big)$ with $\varepsilon_{\text{cal}} = 10^{-4}$, and weighted by channel reliability $r_{e(c)} \in (0,1]$ to give $q(c) = r_{e(c)}\, \tilde{q}(c)$ (channel parameters and reliabilities in Supplementary Section S1).

To prevent double-counting correlated evidence sources, claims for each pair (p, f) are partitioned into correlation groups $\mathcal{G}_{\text{corr}}(p, f)$ by evidence modality (homology, protein-sequence ML, structure, genomic context, genomic language models, inferred context, sequence motifs, localization, and domain-specific predictors; Supplementary Section S1). Stage 0 takes the maximum weighted probability within each group (with an optional second-opinion weight $\beta_{\text{so}} = 0.0$ by default) and pools across independent groups via Noisy-OR:

$$q_g(p,f) = \text{clip}\Big(\max_{c \in g}\big(r_{e(c)}\tilde{q}(c)\big) + \beta_{\text{so}}\, q_{g,(2)}(p,f),\ 10^{-3},\ 0.999\Big),$$

$$P_0(p,f) = \text{clip}\left(1 - \prod_{g \in \mathcal{G}_{\text{corr}}(p,f)} \big(1 - q_g(p,f)\big),\ 10^{-6},\ 1 - 10^{-6}\right). \tag{1}$$

Stage 0 converts $P_0(p,f)$ into the anchored log-odds $L^{(0)}(p,f) = \text{clip}\big(\log \frac{P_0(p,f)}{1-P_0(p,f)}, -12, 12\big)$ and propagates probabilities upward along GO `is_a` and `part_of` edges.

4.3 Stage 1: whole-proteome iterative Bayesian log-odds refinement

Stage 1 refines the proteome-wide probability matrix $\mathbf{P}^{(k)}$ over iterations $k = 1, \ldots, K_{\max}$ by adding six genome-scale prior log-odds shifts $B_m\big(p, f \mid \mathbf{P}^{(k-1)}, \mathcal{K}, \tau_{\text{tax}}, \mathcal{M}, \mathcal{O}\big)$ to the fixed Stage 0 anchor $L^{(0)}(p,f)$ under Jacobi damping:

$$\tilde{L}^{(k)}(p,f) = \text{clip}\left(L^{(0)}(p,f) + \sum_{m=1}^{6} w_m\, B_m\big(p, f \mid \mathbf{P}^{(k-1)}, \mathcal{K}, \tau_{\text{tax}}, \mathcal{M}, \mathcal{O}\right.$$

$$L^{(k)}(p,f) = (1 - \gamma_{\text{iter}})L^{(k-1)}(p,f) + \gamma_{\text{iter}}\tilde{L}^{(k)}(p,f), \qquad P^{(k)}(p,f) = \tag{2}$$

with unit module weights $w_m = 1.0$, damping factor $\gamma_{\text{iter}} = 0.50$, and maximum iterations $K_{\max} = 6$. Iteration terminates when the mean absolute probability change across updated entries falls below $\epsilon = 5 \times 10^{-3}$, and reverts to the last stable iterate if the mean change increases over two consecutive steps. All 18 configurable Stage 1 and Stage 2B hyperparameters were calibrated on *E. coli* K-12 MG1655 and held frozen across all other species (Supplementary Section S2 and Supplementary Table S3). The six priors are defined as follows:

1. *Essentiality prior (B_1, $\alpha_{ess} = 1.50$):* Let $\mathcal{F}_{\text{ess}}(\tau_{\text{tax}})$ be the domain-specific core essential GO profile and $\mathcal{U}_{\text{ess}}(\tau_{\text{tax}})$ the essential categories with $\max_p P^{(0)}(p,e) \le 0.50$. For any term f in the downward subclass closure $\mathcal{D}_{\text{uncov}}(\tau_{\text{tax}}) = \bigcup_{e \in \mathcal{U}_{\text{ess}}(\tau_{\text{tax}})}\{t \in \mathcal{T} : t \sqsubseteq e\}$, candidate claims receive $B_1(p,f) = \alpha_{\text{ess}}\, \mathbb{I}[f \in \mathcal{D}_{\text{uncov}}(\tau_{\text{tax}})]$.
2. *Pathway and heteromeric complex coherence prior (B_2, $\alpha_{coh} = 1.00$):* Missing steps of active GO `has_part` processes receive $w_{\text{proc}}(f) = \max(0, 1 - c_{\text{proc}})$, and missing steps of triggered reference pathways W with step coverage $c_W \in (0,1)$ receive $w_{\text{path}}(f) = \max_{W \ni f} \max(0.25, 1 - c_W)$. For an obligate heteromeric complex $f \in \mathcal{C}_{\text{het}}$ with $N_{\text{strong}}(f)$

proteins at $P^{(0)} > 0.50$ and $N_{\text{cand}}(f)$ candidates at $\max(P^{(0)}, P^{(k-1)}) \geq 0.08$, candidate partners receive $w_{\text{cplx}}(p, f) = 0.85$ when $N_{\text{strong}} = 1 \wedge N_{\text{cand}} \geq 2$ (and 0.5525 when $N_{\text{strong}} \geq 2$), whereas isolated strong singletons ($N_{\text{strong}} = 1 \wedge N_{\text{cand}} \leq 1$) receive -0.75. Therefore, $B_2(p, f) = \alpha_{\text{coh}}\big(\max(w_{\text{proc}}(f), w_{\text{path}}(f)) + w_{\text{cplx}}(p, f)\big)$.

3. *Taxonomic consistency prior (B_3, $\alpha_{cons} = 0.68$):* Active terms $\mathcal{A}^{(k-1)} = \{t \in \mathcal{T} : \max_p P^{(k-1)}(p, t) > 0.50\}$ are checked against direct GO taxon constraints and SAT4J unsatisfiable cores over multi-parent `is_a` and disjointness axioms. Any claim whose term or ancestor lies in the conflict set $\mathcal{V}_{\text{conf}}^{(k-1)}$ receives $B_3(p, f) = -\alpha_{\text{cons}}$.
4. *Anatomical prior (B_4, $\alpha_{anat} = 0.68$, $\alpha_{boost} = 0.65$):* Let $\mathcal{T}_{\text{forb}}^{\text{anat}}(\tau_{\text{tax}})$ denote GO terms forbidden in τ_{tax} through the GO–Uberon/PO/CL bridge (Section 4.4) and $\mathcal{U}_{\text{act}}(\tau_{\text{tax}})$ valid non-root anatomical structures supported by at least two proteins at $P^{(0)} \geq 0.50$. Claims intersecting $\mathcal{T}_{\text{forb}}^{\text{anat}}(\tau_{\text{tax}})$ receive $B_4(p, f) = -0.68$; valid claims linked to $\mathcal{U}_{\text{act}}(\tau_{\text{tax}})$ receive $+0.65$.
5. *Metabolic gap-filling and reaction-local genomic context prior (B_5, $\alpha_{gap} = 1.50$):* Candidate terms matching open metabolic network gaps receive $B_5(p, f) = \alpha_{\text{gap}}$ (1.05 for low-homology gapseq proposals). For each gap reaction R, reaction-local genomic context (RLGC) identifies pathway-neighboring GO terms at 1–2 reaction hops ($\alpha_{\text{rxn}} = 0.50$), weights chromosomal anchor genes p' with $P(p', g) \geq 0.30$ by $w_{\text{anc}}(p', R) = \max_g \alpha_{\text{rxn}}^{d(g)} P(p', g)$, and accumulates Gaussian kernel density $K_{\text{bp}}(\Delta\text{bp}) = \exp(-\Delta\text{bp}^2/(2h_{\text{bp}}^2))$ ($h_{\text{bp}} = 5{,}000$ bp):

$$D_{\text{RLGC}}(p, R) = \sum_{p' : |\Delta\text{bp}(p,p')| \leq 4h_{\text{bp}}} w_{\text{anc}}(p', R)\, K_{\text{bp}}\big(\Delta\text{bp}(p, p')\big)$$

$$S_{\text{RLGC}}(p, R) = \log D_{\text{RLGC}}(p, R) + \beta_{\text{div}}\text{Div}(p) - \gamma_{\text{com}}\text{Comm}(p) - \delta_{\text{self}} P(p, f) + \varepsilon_{\text{str}}\text{Strand}(p) - \zeta_{\text{gap}}\log_{10}(\max(1, \Delta\text{bp}_{\min}(p))) \tag{3}$$

with neighbor-reaction diversity weight $\beta_{\text{div}} = 1.0$, commitment penalty $\gamma_{\text{com}} = 1.0$, self-annotation penalty $\delta_{\text{self}} = 2.0$, same-strand bonus $\varepsilon_{\text{str}} = 0.2$, and minimum-distance penalty $\zeta_{\text{gap}} = 0.1$. In LRO enzyme attribution, S_{RLGC} is combined with orthogroup co-occurrence, Pfam domain compatibility, and (in Track B) Swiss-Prot ESM-2 reaction-centroid cosine similarity.

6. *Co-directional operon prior (B_6, $\alpha_{ctx} = 1.00$, $\alpha_{gap_ctx} = 1.50$, $w_{op_cplx} = 0.95$):* Within a co-directional operon O ($|O| \geq 2$) having active pathway fraction $c_O(W) = k_O(W)/|O|$ for pathway W, a candidate (p, f) receives $B_6(p, f) = \max_{W \ni f : k_O(W) \geq 1}\big(\alpha_{\text{ctx}} c_O(W) \alpha_{\text{gap_ctx}}^{\mathbb{I}[f \in \mathcal{G}_{\text{gap}}]}\big) + w_{\text{op_cplx}}\mathbb{I}[f \in \mathcal{C}_{\text{het}} \wedge \exists p' \in O \setminus \{p\} : P^{(k-1)}(p', f) > 0.50]$.

4.4 Cross-ontology taxonomic and anatomical constraint bridge

To enforce anatomical and taxonomic consistency across domains, we extracted five programmatic constraint tables from the official OBO Foundry releases of GO, Uberon [9], Plant Ontology (PO) [49], Cell Ontology (CL) [50], and NCBI Taxonomy (Supplementary Section S3 and Supplementary Table S7): (1) 26,911 direct GO taxon constraints (13,819 `only_in_taxon` and 13,092 `never_in_taxon`) across 18,159 GO terms [8]; (2) 2,715 cross-ontology equivalence and subclass axioms linking 2,674 GO terms to 1,294 Uberon, PO, and CL classes across 30 relations (including `results_in_development_of`, `results_in_morphogenesis_of`, `results_in_formation_of`, `results_in_maturation_of`, and `occurs_in`); (3) 8,331 taxon constraints (4,767 `only_in_taxon` and 3,564 `never_in_taxon`) over 1,247 Uberon, PO, and CL anatomical classes; (4) 7,722 transitive `is_a` and `part_of` edges over 2,855 anatomical classes; and (5) 1,818 taxonomic lineage edges over 1,305 NCBI Taxonomy clades. A GO term t is entailed as anatomically forbidden in taxon τ_{tax} whenever t (or a GO ancestor of t) links to an anatomical class u whose anatomical ancestor closure carries an `only_in_taxon` or `never_in_taxon` rule incompatible with the taxonomic lineage of τ_{tax}.

4.5 Stage 2A: deterministic symbolic enforcement and semantic ancestor relaxation

After Stage 1 convergence, Stage 2A deterministically enforces all direct GO taxon constraints, cross-ontology anatomical constraints, and SAT4J logical consistency axioms prior to final upward GO ancestor closure, ensuring that clade-incompatible leaf claims are resolved before parent terms are populated. For each protein p and forbidden term t, Stage 2A traverses $\text{Anc}(t)$ (excluding the three GO aspect roots) to identify the minimal valid ancestor set $\mathcal{A}_{\text{valid}}(t, \tau_{\text{tax}})$ comprising the most specific ancestors of t that are both taxonomically and anatomically admissible in τ_{tax}. Reconstructing the unpenalized pre-conflict probability $P_{\text{unpen}}(p, t)$ from $L^{(0)}(p, t)$ and positive Stage 1 priors, semantic ancestor relaxation transfers this support to each minimal valid ancestor and zeroes the forbidden term:

$$P_{\text{Stage2A}}(p, a) = \max\Big(P^{(K)}(p, a),\, \gamma_{\text{relax}}\, P_{\text{unpen}}(p, t)\Big), \qquad \forall a \in \mathcal{A}_{\text{valid}}$$
$$P_{\text{Stage2A}}(p, t) = 0, \tag{4}$$

with $\gamma_{\text{relax}} = 1.0$, followed by upward true-path ancestor closure over admissible terms. In addition, for any obligate heteromeric complex $f \in \mathcal{C}_{\text{het}}$ with a single protein at $P \geq 0.50$ and zero partner candidates at $P \geq 0.08$ across the genome, Stage 2A attenuates the isolated singleton to 0.45.

4.6 Stage 2B: role-aware reaction support and positive-cost lexicographic pFBA

Stage 2B operates over a sanitized universal metabolic network $\mathcal{U}_{\text{clean}}$ derived from ModelSEED [43] (47,880 reactions and 33,978 metabolites, of which 14,243 carry EC mappings and 11,826 are mass- and charge-balanced enzymatic reactions; 2,417 pool or elementally imbalanced reactions receive penalty $w_{\text{imb}}(j) = 100.0$). Each reaction j is associated with a disjunctive normal form (DNF) catalytic rule $\mathcal{M}_{\text{dnf}}(j) = \{m_1, \ldots, m_{K_j}\}$ over alternative isozyme mechanisms m, where each mechanism requires a conjunction of subunit roles $\mathcal{R}_{\text{sub}}(m) = \{u_1, \ldots, u_{D_m}\}$. Given candidate locus probabilities $s_{p,u} \in (0, 1)$, Stage 2B aggregates the top $M_{\text{iso}} = 3$ candidates per role via Noisy-OR, combines required subunits via a bottleneck-gated geometric mean at $\theta_{\text{sub}} = 0.25$, and takes the maximum across isozyme mechanisms:

$$Q(u) = 1 - \prod_{p \in \text{Top-}M_{\text{iso}}(\mathcal{C}_{\text{cand}}(u))} \big(1 - s_{p,u}\big),$$

$$S_{\text{mech}}(m) = \begin{cases} \Big(\prod_{u \in \mathcal{R}_{\text{sub}}(m)} Q(u)\Big)^{1/D_m}, & \min_u Q(u) \geq \theta_{\text{sub}}, \\ \min_{u \in \mathcal{R}_{\text{sub}}(m)} Q(u), & \text{otherwise}, \end{cases} \tag{5}$$

Reactions with $S_{\text{rxn}}(j) \geq \tau_{\text{core}} = 0.50$, together with open medium exchanges and spontaneous reactions, form the draft core $\mathcal{R}_{\text{core}}$. Candidate repair reactions $j \in \mathcal{D}_{\text{dom}} \setminus \mathcal{R}_{\text{core}}$ receive a strictly positive addition cost $c_j^{\text{add}} = \lambda_{\text{add}}(1 - S_{\text{rxn}}(j)) + w_{\text{imb}}(j) + \epsilon_{\text{par}}$ ($\lambda_{\text{add}} = 1.0$, $\epsilon_{\text{par}} = 10^{-3}$). Splitting reversible fluxes into non-negative components $v_j = v_j^+ - v_j^-$ and setting the biomass target on medium $\mathcal{E}_{\text{med}}$ to $v_{\text{bio}}^{\text{target}}(\mathcal{E}_{\text{med}}) = \max(\alpha_{\text{bio}} v_{\text{bio}}^{\max}(\mathcal{E}_{\text{med}}), 0.1)$ ($\alpha_{\text{bio}} = 0.10$), Stage 2B solves a two-step lexicographic pFBA [53] using the HiGHS linear programming solver (feasibility tolerance 10^{-9}). Step 1 minimizes the repair cost $\sum_{j \notin \mathcal{R}_{\text{core}}} c_j^{\text{add}}(v_j^+ + v_j^-)$ subject to $\mathbf{Sv} = \mathbf{0}$ and $v_{\text{bio}} \geq v_{\text{bio}}^{\text{target}}(\mathcal{E}_{\text{med}})$ (solving first over balanced reactions and admitting penalized imbalanced reactions only if infeasible) to obtain optimal repair cost $C^*(\mathcal{E}_{\text{med}})$. Step 2 minimizes total non-biomass flux $\sum_{j \neq \text{bio}}(v_j^+ + v_j^-)$ subject to $\sum_{j \notin \mathcal{R}_{\text{core}}} c_j^{\text{add}}(v_j^+ + v_j^-) \leq (1 + 10^{-6})C^*(\mathcal{E}_{\text{med}}) + 10^{-9}$. The active repair set $\mathcal{J}_{\text{rep}}(\mathcal{E}_{\text{med}}) = \{j \notin \mathcal{R}_{\text{core}} : |v_j^{\text{pFBA}}(\mathcal{E}_{\text{med}})| > 10^{-6}\}$ is evaluated by flux-variability ($1.01C^*$), reaction-knockout ($1.10C^*$), and five-replicate cost-perturbation ($\pm 5\%$) certificates to yield the flux confidence $\pi_j^{\text{FBA}} = 0.5\mathbb{I}[j \in \mathcal{R}_{\text{FVA}}] + 0.3\mathbb{I}[j \in \mathcal{R}_{\text{KO}}] + 0.2\omega_{\text{ens}}(j) \in [0, 1]$ (Supplementary Section S4 and Supplementary Table S12).

4.7 Stage 2B: competitive gene-locus responsibility and substrate probing

For each repaired reaction $j \in \mathcal{J}_{\text{rep}}$ and required role $u \in \mathcal{R}_{\text{sub}}(m_j^*)$, Stage 2B distributes locus responsibility competitively across candidate genes $p \in \mathcal{C}_{\text{cand}}(u)$ and an explicit unassigned-gene hypothesis $h_\varnothing$ with prior mass $\eta_{\text{null}} = 0.05$:

$$w_{j,u}(p) = \big(o_{p,u}\big)^{\beta_1} \text{BF}_{\text{operon}}(p, u)^{\beta_2} \exp\big(\beta_3 C_{\text{syn}}(p, u) + \beta_4 S_{\text{ESM2}}(p, u)$$

$$r_{j,u}(p) = \frac{w_{j,u}(p)}{\eta_{\text{null}} + \sum_{p' \in \mathcal{C}_{\text{cand}}(u)} w_{j,u}(p')}, \tag{6}$$

where $o_{p,u} = s_{p,u}/(1 - s_{p,u})$ is the sequence odds, $\text{BF}_{\text{operon}}(p, u)$ is the RLGC operon Bayes factor, $C_{\text{syn}}(p, u)$ is the operon and pathway synteny score, $S_{\text{ESM2}}(p, u)$ is the language-model similarity score, and $\beta_1 = \beta_2 = 1.0, \beta_3 = \beta_4 = 1.5$. The backward log-odds update to locus p is $\Delta L_{\text{FBA}}(p, u) = \gamma_{\text{FBA}} \sum_{j \in \mathcal{J}_{\text{rep}}(u)} \pi_j^{\text{FBA}} r_{j,u}(p)(1 - S_{\text{rxn}}(j))$ with $\gamma_{\text{FBA}} = 2.20$, applied across damped outer iterations $\lambda_t = \lambda_0 \gamma_{\text{damp}}^{t-1}$ ($\lambda_0 = 0.70, \gamma_{\text{damp}} = 0.60$). For catabolic enzymes in Price-149, Stage 2B supplements minimal-glucose pFBA with cognate-substrate stoichiometric neighborhood probing (rewarding candidate reactions that share non-currency metabolites at 1–2 reaction hops with chromosomal neighbors within ±10 genes) and blends single-predictor and five-predictor ensemble posteriors (complete equations in Supplementary Section S4). Held-out metabolic reactions are classified by five binary recoverability indicators: structural presence and mass balance in $\mathcal{U}_{\text{clean}}$ ($\mathcal{C}_{\text{struct}}$), medium-specific flux demand ($\mathcal{C}_{\text{task}}$, partitioning core biomass Stratum A, cognate-substrate catabolic Stratum B, and secondary Stratum C), isozyme uniqueness ($\mathcal{C}_{\text{unique}}$), operon support ($\mathcal{C}_{\text{operon}}$), and multi-subunit partner presence ($\mathcal{C}_{\text{complex}}$).

$$S_{\text{gold}}(j) = \max_{m \in \mathcal{M}_{\text{dnf}}(j)} S_{\text{mech}}(m)$$

4.8 Stage 3: selective neuro-symbolic annotation completion

Stage 3 is triggered selectively on four post-Stage-2 anomaly categories: (1) pathway gaps (pathway completeness ≥ 0.50 with candidate posteriors in $[0.05, 0.50)$; ≥ 0.65 and $[0.15, 0.49]$ in the controlled 120-anomaly benchmark); (2) incomplete obligate heteromeric complexes (one active subunit at $P \geq 0.50$ and sub-threshold partner candidates); (3) borderline uncovered essential categories; and (4) borderline taxonomic or anatomical conflicts ($P \in [0.35, 0.62]$) amenable to ancestral GO relaxation. Stage 3 evaluates at most 8 candidates per anomaly and 150 anomalies per genome.

Each anomaly is stripped of evaluation labels and gene symbols and processed in a bounded multi-turn tool-calling loop (maximum 6 turns; 2 turns in the fast mode) by a locally hosted FP8-quantized 27-billion-parameter instruction model from the Qwen open-weight family [71] served via vLLM (32,768-token context, temperature 0.0, maximum 450 output tokens per turn, evaluated across seeds 42, 43, and 44). The model invokes five deterministic read-only tools to inspect the operon neighborhood, inspect the pathway or complex gap, query protein evidence and domain signatures, check symbolic ontology constraints, and propose a structured resolution (PROMOTE, RELAX, DEMOTE, or ABSTAIN). Every proposal is verified by the

deterministic symbolic guardrail against the 48,101 GO term stanzas, 1,305 NCBI Taxonomy clades, 26,911 direct GO taxon rules, and 2,715 GO–Uberon/PO/CL axioms before acceptance (tool schemas, de-identification protocol, and per-seed results in Supplementary Section S6 and Supplementary Table S14).

4.9 Experimental phenotype prediction in BacDive

We evaluated organism-level phenotype prediction across 2,332 carbon-source utilization assays in 529 ProTraits-matched bacterial strains, 17,496 diagnostic enzyme assays in 3,017 BacDive type strains, and 50,075 assays for 10 diagnostic enzymes across 6,313 strains [60, 61]. Given binary pathway or reaction calls $g(s,y), m(s,y), c(s,y) \in \{0,1\}$ from gapseq v1.2 [45], ModelSEED v2.6.1 [43], and CarveMe v1.5.2 [44] for strain s and assay y, we form the blended matrix $B(s,y)$ ($0.55g + 0.30m + 0.15c$ for carbon; $0.55g + 0.28m + 0.17c$ for enzymes), column-center each assay $\tilde{B}_{s,y} = B(s,y) - \bar{B}_y$, and compute the row-normalized non-negative cosine coupling matrix $\Psi(y,y')$ with zero diagonal to obtain the unsupervised cross-assay coupling score $\psi(s,y) = \sum_{y'} \Psi(y,y')(B(s,y') - 0.5)$. In blind Track 1 (0% phenotype labels), binary tool calls are mapped to calibrated probabilities $p_{\text{gap}}, p_{\text{ms}}, p_{\text{cm}}$ and pooled in log-odds space:

$$P_{\text{Track1}}(s,y) = \sigma_{\log}\Big(w_{\text{gap}} \operatorname{logit}\big(p_{\text{gap}}(s,y)\big) + w_{\text{ms}} \operatorname{logit}\big(p_{\text{ms}}(s,y)\big) + w_{\text{cm}} \operatorname{logit}\big(p_{\text{cm}}(s,y)\big) + w_{\psi}\, \psi(s,y) \Big) \tag{7}$$

with weights $(0.56, 0.28, 0.16, 0.65)$ for carbon and $(0.55, 0.27, 0.18, 0.55)$ for enzymes at threshold 0.50. Supervised baselines (Traitar L1-SVM [63], BacDive Random Forest [61], and DiASPora GBDT [62]) and the GSPA + DiASPora hybrid were evaluated under five-fold genus-blocked cross-validation with all calibration, prevalence, and coupling parameters estimated strictly within training folds. In Track 2, 30% of each strain's assays are revealed to condition the Bayesian anchor $P_{\text{Track2}}(s,y)$ on revealed substrate co-occurrence or genus-level EC posteriors, and predictions are evaluated on the held-out 70% (complete parameterization in Supplementary Section S5 and Supplementary Table S13).

4.10 Benchmarks, time-based and homology-based split retraining, metrics, and structural audit score

To evaluate prospective generalization across time and across sequence-similarity regimes on complete genome inputs, we constructed a six-split whole-genome withheld CAFA benchmark across six complete prokaryotic reference proteomes (*B. subtilis* 168, *P. aeruginosa* PAO1, *M. tuberculosis* H37Rv, *H. pylori* 26695, *Synechocystis* sp. PCC 6803, and *M. jannaschii* DSM 2661; 20,697 coding sequences, 3,315 experimentally annotated target proteins, 1,060 strictly post-t_0 targets, 3,809 operons, and 1,236 metabolic gaps) alongside a multi-clade isolated-protein CAFA benchmark (Table 1 and Supplementary Tables S4 and S15). In the two whole-genome time-based splits, post-t_0 targets (1,060 proteins) are evaluated either with pre-t_0 experimental annotations on the six chromosomes retained in training to test chromosomal rollover context (`WG_Time_Rollover`, 82,404 training proteins) or with all 2,524 experimentally annotated proteins from the six species purged from training (`WG_Time_Withheld`, 79,880 training proteins). In the four whole-genome homology splits, all six species are purged from training and evaluated across all 3,315 experimental target proteins after additionally purging all cross-species training homologs sharing pairwise sequence identity $> \eta_{\text{id}}$ with any protein in the six withheld genomes (`WG_Homology_le100`: 79,880 train; `WG_Homology_le70`: 79,457 train; `WG_Homology_le50`: 76,405 train; `WG_Homology_le30`: 61,323 train). On the two time-based splits, we additionally evaluate the official pre-trained weights of DeepFRI [27] and DeepGO-SE [25]. On every split, all 14 trainable per-protein baselines (term-frequency prior, filtered DIAMOND database, L_2-regularized InterPro domain logistic regression, canonical 16-kernel 1D convolutional DeepGOCNN with $k \in \{8, 16, \ldots, 128\} \times 512$ filters = 8,192 features, DeepGOPlus [19], 2,560-dimensional ESM-2 3B `esm2_t36_3B_UR50D` k-NN index [72], ESM-2 3B MLP, hierarchical ESM-2 3B C-HMCNN with max-constraint loss [73], DeepFRI graph convolutional and language-model architecture [27], DeepGraphGO two-layer graph convolutional network over the DIAMOND sequence-similarity graph with InterPro node features [20], TALE+ transformer encoder with hierarchical GO label-embedding joint attention and DIAMOND fusion [21], SPROF-GO language-model self-attention pooling with hierarchical label diffusion and homology transfer [24], DeepGO-SE five-model approximate semantic entailment ensemble over $\mathcal{EL}^{++}$ description-logic embeddings [25], and DPFunc InterPro domain cross-attention with graph convolutional layers [29]) together with our out-of-fold GSPA-Seq meta-stacker are retrained from scratch on that split's filtered training set before running whole-genome inference across all 20,697 coding sequences in GSPA Stages 0–2. For the pan-phylum AllTheBacteria and reference eukaryote analysis (72,533 complete genomes across 11,632 species and 37 phyla [54]; Supplementary Section S7 and Supplementary Table S17), coding sequences across 73 assembly batches (315,671,487 proteins) were deduplicated by SHA-256 sequence hashing (73,866,871 unique sequences) and clustered via `MMseqs2 linclust` [64] at 90% sequence identity and 80% coverage (36,804,001 non-redundant cluster representatives), scored with our full GSPA-Seq stacker, mapped back to each chromosome's co-directional operon layout, and integrated in shared-ontology multi-genome batch mode (`gspa-unified.jar integrate -batch-manifest`). Across whole-proteome benchmarks, donor-homology masking removes reference proteins sharing $\text{pident}(g,r) > \eta_{\text{id}}$ ($\eta_{\text{id}} \in \{100\%, 70\%, 50\%, 30\%\}$) or belonging to the test organ-

ism's genus or family across DIAMOND databases, ESM-2 and ProtT5 neighbor indices, FoldSeek structural templates, and calibration sets. In leave-reaction-out benchmarks, Track A masks sequence alignments and embedding donors with cosine similarity > 0.50 without reaction centroids, while Track B masks sequence alignments and retains Swiss-Prot ESM-2 reaction centroids.

We report official CAFA information-accretion-weighted f_w ($WF_{\max}$), protein-centric unweighted $F_{\max}$, pair-level micro $F_{\max}$, and minimum semantic distance $S_{\min}$ [1, 33, 74] over threshold grids $\theta \in \{0.01, \ldots, 0.99\}$ ($\{0.10, \ldots, 0.90\}$ in the eukaryotic stress test), alongside Hit@k and mean reciprocal rank (MRR) for candidate ranking, 95% bootstrap or exact Clopper–Pearson confidence intervals, and two-sided paired Wilcoxon signed-rank or exact binomial McNemar tests (Supplementary Section S5). All CPU benchmarks ran on dual-socket AMD EPYC 7763 nodes (8 threads per genome), with PLM, 16-kernel DeepGOCNN, ESM-2 3B, external SOTA GO baseline retraining, and Stage 3 LLM inference executed on NVIDIA A100-SXM4-80GB, NVIDIA H100-80GB, and NVIDIA H200-140GB GPUs.

The Genome-Scale Annotation Evaluation Framework (GAEF) summarizes structural proteome invariants at $\theta_{\text{eval}} = 0.50$ by combining essential function completeness C_{ess} ($w_{\text{ess}} = 0.30$), functional pathway/complex/process coherence C_{func} ($w_{\text{func}} = 0.40$), taxonomic and anatomical consistency $C_{\text{cons}} = \max(0, 1 - 0.01V(\mathcal{P}))$ ($w_{\text{cons}} = 0.30$), and functional resolution $C_{\text{res}} = \min(1, \text{meanIC}/10)$ ($w_{\text{res}} = 0.0$):

$$\text{GAEF}(\mathcal{P}) = \frac{\sum_{d \in \{\text{ess,func,cons,res}\}: C_d \neq \text{null}} w_d C_d}{\sum_{d: C_d \neq \text{null}} w_d}, \quad (8)$$

which equals $0.30C_{\text{ess}} + 0.40C_{\text{func}} + 0.30C_{\text{cons}}$ when all three structural dimensions are active. Because GAEF measures the structural invariants targeted by Stages 1 and 2, we report it as an internal structural audit score alongside independent held-out functional and experimental phenotype evaluations.

Data availability

All precomputed cross-ontology constraint tables and hierarchies (Supplementary Table S7), obligate heteromeric complex classifications, benchmark manifests with cryptographic checksums, frozen evaluation tables underlying Figures 1–6 and Tables 1–8, and a complete provenance map linking every manuscript value to its underlying result table are available under CC-BY-4.0 at https://github.com/bio-ontology-research-group/gspa (release v2.1.0).

Code availability

The GSPA v2.1.0 software suite, including the command-line integration and evaluation engine, the Stage 2B stoichiometric locus refiner, the Stage 3 neuro-symbolic agent and ontology guardrail, and all benchmark and figure generation scripts, is available under the GPL-3.0-or-later license at https://github.com/bio-ontology-research-group/gspa (release v2.1.0).

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Author contributions

T.A.-S., M.K., and R.H. conceived the study and designed the methodology. T.A.-S. and R.H. implemented the software, cross-ontology bridges, stoichiometric refiner, and benchmark suites. All authors analyzed the results and wrote the manuscript.

Competing interests

The authors declare no competing interests.

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